

Installing Tableau Server

in a Linux Environment on Google Cloud Platform

by Mohamad Ikhsan Nurulloh



Google Cloud

Disclaimer

- **No Endorsement:** I am not endorsed by any provider; I am sharing this information purely for fun and educational purposes.
- **Testing and Development Only:** This content is intended solely for testing and development purposes.
- **Refer to Official Tableau Documentation:** For production environments, always refer to the official Tableau documentation to ensure compliance with best practices and guidelines.
- **Proceed at Your Own Risk:** If you choose to follow this guide, you do so at your own risk. Ensure that you understand the implications of deploying software and services in a cloud environment.

Overview of Installing Tableau Server on GCP



Installing **Tableau Server** on **Google Cloud Platform (GCP)** provides organizations with a scalable and flexible solution for data visualization and analytics. By creating a Google Compute Engine (GCE) virtual machine (VM) with a compatible Linux distribution, users can leverage cloud capabilities while avoiding the overhead of physical servers. Key benefits include easy scalability, integration with Google Cloud services, and high availability, enabling organizations to enhance their data-driven decision-making in a robust cloud environment.

Before Installing Tableau Server on GCP

1 . Preparation

- Validating Your Server Deployment Plan

Consider Tableau Cloud for better performance. If self-hosting or using cloud provider, refer to the [Tableau Enterprise Deployment Guide](#) (EDG).

- Hardware Recommendations for Production Installations

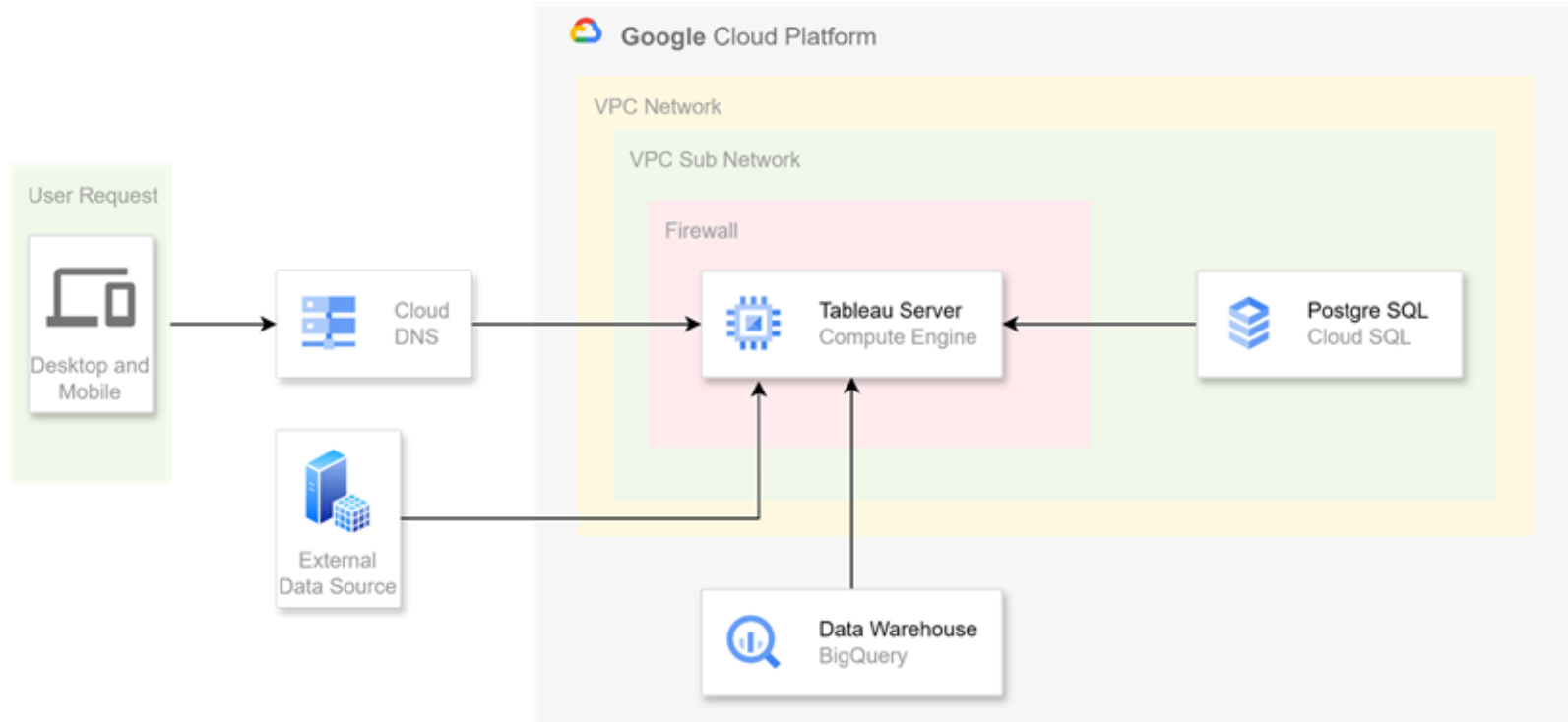
Check the EDG for optimal hardware specifications for CPU, memory, and storage.

- Choosing the Tableau Server Version

Use the latest stable version for new features and security.

2 . Tableau Server Architecture

Architecture: Single Node Tableau Server



3 . Tableau Server Architecture Explanation

1. User Request:

- Desktop and Mobile: Users access Tableau Server from various devices.

2. Google Cloud Platform (GCP):

- VPC Network: Secure environment.
- VPC Sub Network: Hosting the Tableau Server.
- Firewall: Controls and monitors network traffic.

3. Tableau Server:

- Compute Engine: Instance running the Tableau Server for processing and visualization tasks.

4. Data Sources:

- PostgreSQL (Cloud SQL): Managed database for storing data.
- Data Warehouse (BigQuery): Scalable storage and fast querying capabilities.
- External Data Source: Additional data that can be integrated with Tableau Server.

5. Cloud DNS:

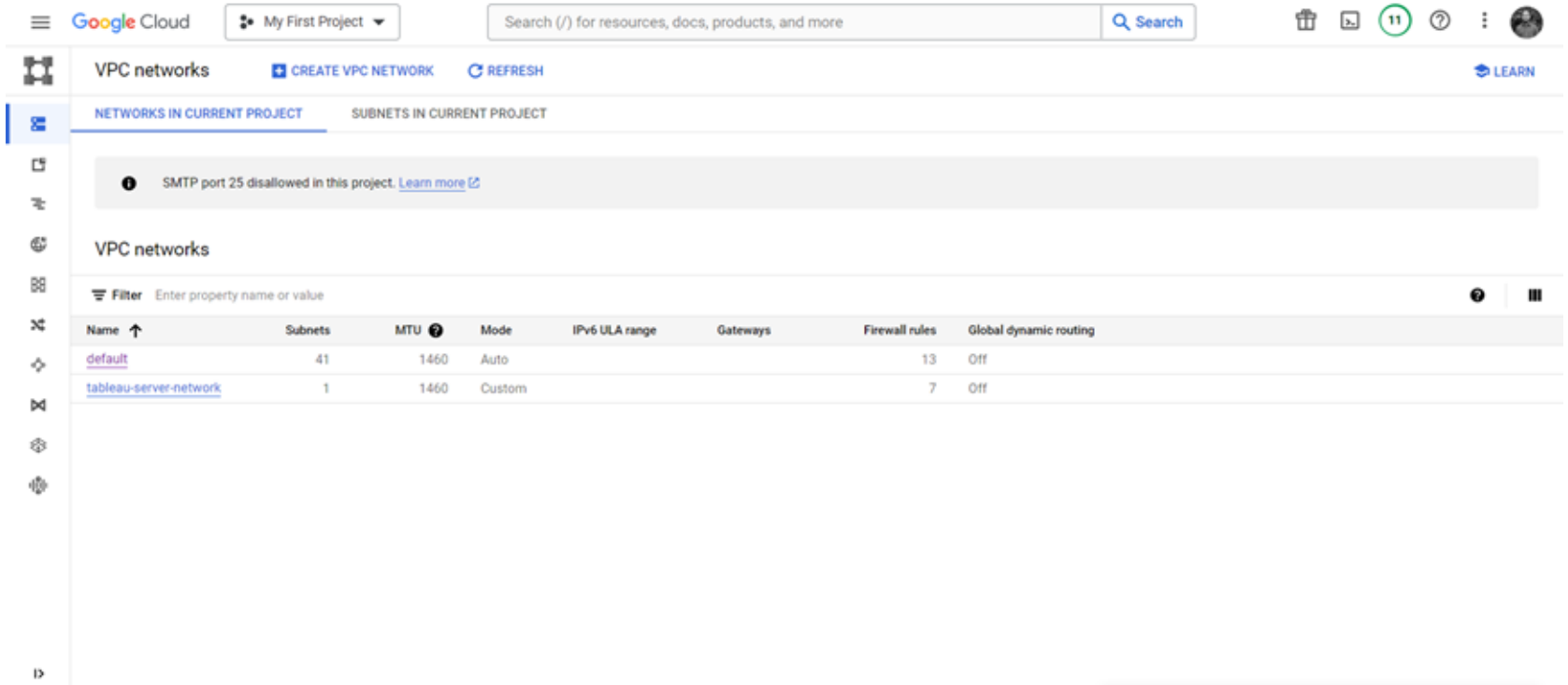
- Manages DNS configurations for accessing the Tableau Server.

4 . Tableau Server Architecture Workflow

1. **User Request Handling:** Users send requests from their devices via Cloud DNS to the Tableau Server.
2. **Data Processing and Visualization:** Tableau Server processes requests and interacts with PostgreSQL and BigQuery to fetch and analyze data.
3. **External Data Integration:** Tableau Server can also access and utilize additional external data sources.
4. **Firewall Security:** The firewall ensures only authorized traffic reaches the Tableau Server within the VPC network.

Installing Tableau Server on GCP

1 . Creating VPC



The screenshot shows the Google Cloud console interface for VPC networks. At the top, there's a header with the Google Cloud logo, a project selector set to 'My First Project', a search bar, and user account icons. Below the header, the 'VPC networks' section is active, with buttons for 'CREATE VPC NETWORK' and 'REFRESH'. A message banner indicates that SMTP port 25 is disallowed in this project. The main content area displays a table of VPC networks for the current project.

Name ↑	Subnets	MTU ⓘ	Mode	IPv6 ULA range	Gateways	Firewall rules	Global dynamic routing
default	41	1460	Auto			13	Off
tableau-server-network	1	1460	Custom			7	Off

Establish a Virtual Private Cloud (VPC) to host Tableau Server, ensuring a secure and isolated network environment.

2 . Creating Subnet

The screenshot shows the Google Cloud console interface for a VPC network named 'tableau-server-network'. The 'SUBNETS' tab is selected, displaying a table of existing subnets. Below the table, there is a section for 'Reserved proxy-only subnets for load balancing' which is currently empty. The interface includes a search bar at the top, navigation tabs for various network settings, and a sidebar with icons for different cloud services.

Google Cloud My First Project Search (/) for resources, docs, products, and more Search

VPC network details DELETE VPC NETWORK SHOW INFO PANEL

tableau-server-network

OVERVIEW SUBNETS STATIC INTERNAL IP ADDRESSES FIREWALLS FIREWALL ENDPOINTS ROUTES VPC NETWORK PEERING PRIVATE SERVICES ACCESS DNS CONF >

Subnets ADD SUBNET FLOW LOGS

Filter Enter property name or value

☐	Name ↑	Region	Stack Type	Primary IPv4 range	Secondary IPv4 ranges	IPv6 ranges	Reserved internal ranges	Gateway	Private Google Access	Flow logs ?
☐	tableau-server-subnet	asia-southeast2	IPv4	10.184.0.0/20			None	10.184.0.1	Off	Off

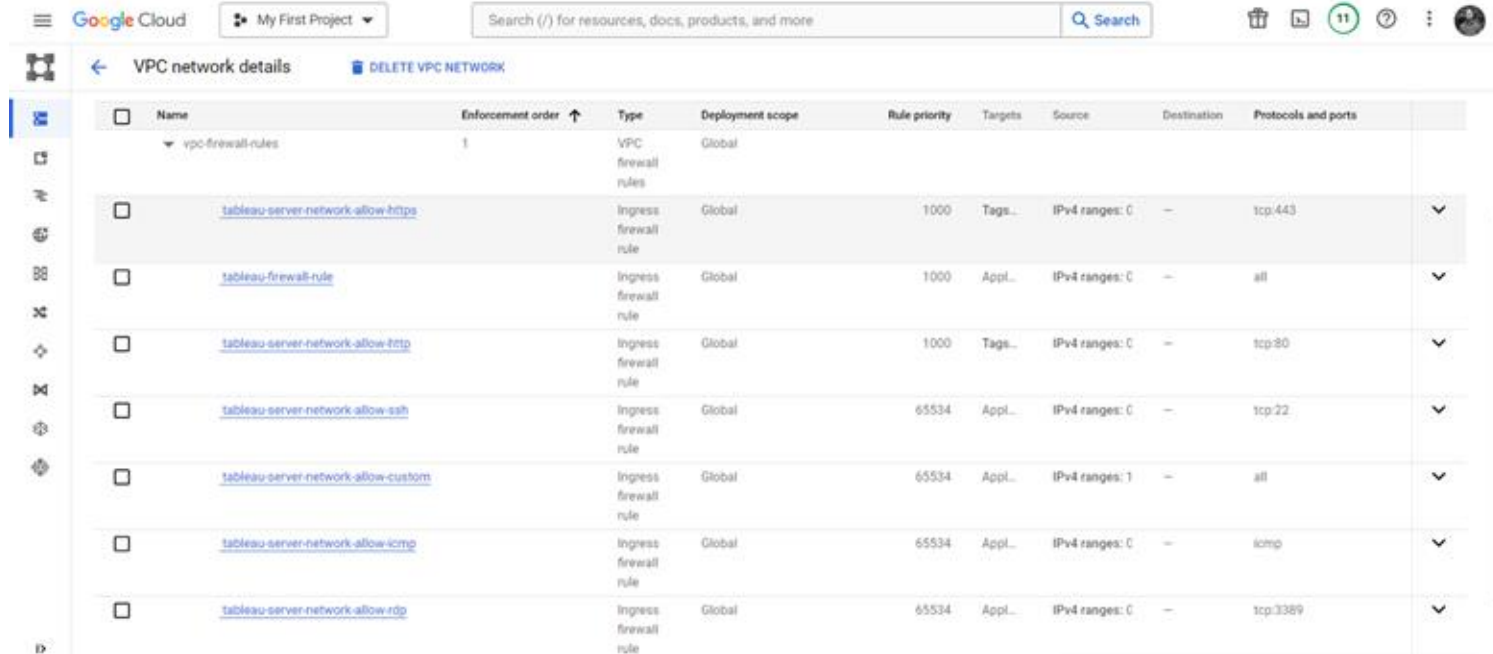
Reserved proxy-only subnets for load balancing

☐	Name	Region ↑	IP address ranges	Gateway	Role	Purpose
No rows to display						

EQUIVALENT REST

Set up a subnet within the VPC to manage network traffic efficiently and segment the network for better security.

3 . Creating Firewall Rule



The screenshot shows the Google Cloud console interface for VPC network details. The page title is "VPC network details" with a "DELETE VPC NETWORK" button. A table lists the firewall rules. The table has columns: Name, Enforcement order, Type, Deployment scope, Rule priority, Targets, Source, Destination, and Protocols and ports. The rules are for Tableau Server, allowing access to various ports like 443, 80, 22, and 3389.

Name	Enforcement order	Type	Deployment scope	Rule priority	Targets	Source	Destination	Protocols and ports
vpc-firewall-rules	1	VPC firewall rules	Global					
tableau-server-network-allow-https		Ingress firewall rule	Global	1000	Tags...	IPv4 ranges: C	—	tcp:443
tableau-firewall-rule		Ingress firewall rule	Global	1000	Appl...	IPv4 ranges: C	—	all
tableau-server-network-allow-http		Ingress firewall rule	Global	1000	Tags...	IPv4 ranges: C	—	tcp:80
tableau-server-network-allow-ssh		Ingress firewall rule	Global	65534	Appl...	IPv4 ranges: C	—	tcp:22
tableau-server-network-allow-custom		Ingress firewall rule	Global	65534	Appl...	IPv4 ranges: 1	—	all
tableau-server-network-allow-icmp		Ingress firewall rule	Global	65534	Appl...	IPv4 ranges: C	—	icmp
tableau-server-network-allow-rdp		Ingress firewall rule	Global	65534	Appl...	IPv4 ranges: C	—	tcp:3389

Create firewall rules to control access to the Tableau Server, ensuring that only authorized users can connect.

<https://help.tableau.com/current/server/en-us/ports.htm>

https://help.tableau.com/current/desktopdeploy/en-us/desktop_deploy_proxy.htm

4 . Creating Compute Engine VM

a . Machine Configuration

Create an instance [CREATE VM FROM...](#) [EQUIVALENT CODE](#)

New VM instance
Create a single VM instance from scratch

New VM instance from template
Create a single VM instance from an existing template

New VM instance from machine image
Create a single VM instance from an existing machine image

Marketplace
Deploy a ready-to-go solution onto a VM instance

Name *
tableau-server-test

MANAGE TAGS AND LABELS

Region *
asia-southeast2 (Jakarta)
Region is permanent

Zone *
Any
Google will choose a zone on your behalf, maximizing VM obtainability. Zone is permanent.

Machine configuration

☒ General purpose ☐ Compute optimized ☐ Memory optimized ☐ Storage optimized ☐ GPUs

Machine types for common workloads, optimized for cost and flexibility

Series	Description	vCPUs	Memory	Platform
☐ C4	PREVIEW Consistently high performance	2 - 192	4 - 1,488 GB	Intel Emerald Rapids
☐ N4	Flexible & cost-optimized	2 - 80	4 - 640 GB	Intel Emerald Rapids
☐ C3	Consistently high performance	4 - 192	8 - 1,536 GB	Intel Sapphire Rapids
☐ C3D	Consistently high performance	4 - 360	8 - 2,880 GB	AMD Genoa
☒ E2	Low cost, day-to-day computing	0.25 - 32	1 - 128 GB	Based on availability
☐ N2	Balanced price & performance	2 - 128	2 - 864 GB	Intel Cascade and Ice L

Monthly estimate
\$34.19
That's about \$0.05 hourly
Pay for what you use: no upfront costs and per second billing

Item	Monthly estimate
2 vCPU + 4 GB memory	\$32.89
10 GB balanced persistent disk	\$1.30
Total	\$34.19

[Compute Engine pricing](#)
[LESS](#)

[CREATE](#) [CANCEL](#) [EQUIVALENT CODE](#)

Select an appropriate machine type based on your expected workload and performance needs .

4 . Creating Compute Engine VM

b . Custom Machine Type

The screenshot shows the Google Cloud 'Create an instance' page. The left sidebar lists options: 'New VM instance' (selected), 'New VM instance from template', 'New VM instance from machine image', and 'Marketplace'. The main area is titled 'Machine type' and includes a 'CREATE VM FROM...' button. Below this, there are 'PRESET' and 'CUSTOM' tabs, with 'CUSTOM' selected. A note states: 'Creating a custom machine incurs additional costs'. The configuration section shows 'Cores' set to 8 (via a slider from 2 to 32) and 'Memory' set to 64 GB (via a slider from 4 to 64). A 'Shared core' checkbox is unchecked. Below this is the 'ADVANCED CONFIGURATIONS' section, which includes 'Availability policies' with a dropdown for 'VM provisioning model' set to 'Standard'. At the bottom are 'CREATE', 'CANCEL', and 'EQUIVALENT CODE' buttons. On the right, a 'Monthly estimate' section shows a total of \$356.23, broken down into vCPU + memory (\$354.93) and disk (\$1.30).

Google Cloud My First Project Search (/) for resources, docs, products, and more Search

Create an instance CREATE VM FROM... EQUIVALENT CODE

New VM instance
Create a single VM instance from scratch

New VM instance from template
Create a single VM instance from an existing template

New VM instance from machine image
Create a single VM instance from an existing machine image

Marketplace
Deploy a ready-to-go solution onto a VM instance

Machine type
Choose a machine type with preset amounts of vCPUs and memory that suit most workloads. Or, you can create a custom machine for your workload's particular needs. [Learn more](#)

PRESET CUSTOM

Creating a custom machine incurs additional costs

Cores 2 32 8 vCPU (4 core)

☐ Shared core

Memory 4 64 64 GB

ADVANCED CONFIGURATIONS

Availability policies

VM provisioning model Standard

Choose "Spot" to get a discounted, preemptible VM. Otherwise, stick to "Standard". [Learn more](#)

VM PROVISIONING MODEL ADVANCED SETTINGS

CREATE CANCEL EQUIVALENT CODE

Monthly estimate
\$356.23
That's about \$0.49 hourly
Pay for what you use: no upfront costs and per second billing

Item	Monthly estimate
8 vCPU + 64 GB memory	\$354.93
10 GB balanced persistent disk	\$1.30
Total	\$356.23

[Compute Engine pricing](#)

LESS

If necessary, configure a custom machine type to tailor resources to specific requirements .

4 . Creating Compute Engine VM

c . Boot disk

The screenshot shows the Google Cloud Platform interface for creating a new VM instance. The 'Create an instance' page is active, with the 'Container' tab selected. The 'Boot disk' section is expanded, showing the following configuration:

Property	Value
Name	tableau-server-test
Type	New balanced persistent disk
Size	100 GB
Snapshot schedule	No schedule selected
License type	Free
Image	Ubuntu 22.04 LTS

The 'Monthly estimate' on the right is \$367.93, which is approximately \$0.50 per hour. The estimate is based on 8 vCPU + 64 GB memory (\$354.93) and 100 GB balanced persistent disk (\$13.00).

The 'Identity and API access' section shows the 'Service account' set to 'Compute Engine default service account'. A note indicates that the 'Service Account User role' must be set for users who want to access VMs with this service account.

At the bottom, there are buttons for 'CREATE', 'CANCEL', and 'EQUIVALENT CODE'.

Choose and configure the boot disk for the VM, ensuring it meets the installation requirements .
https://help.tableau.com/current/server/en-us/server_hardware_min.htm

4 . Creating Compute Engine VM

d . Firewall Configuration

The screenshot shows the Google Cloud 'Create an instance' page. The left sidebar lists options: 'New VM instance' (selected), 'New VM instance from template', 'New VM instance from machine image', and 'Marketplace'. The main content area is for 'New VM instance' and includes a note about Service Account User role, 'Access scopes' (with 'Allow default access' selected), 'Firewall' (with 'Allow HTTP traffic' and 'Allow HTTPS traffic' checked), 'Observability - Ops Agent' (with 'Install Ops Agent for Monitoring and Logging' unchecked), and 'Advanced options'. A 'Monthly estimate' table is on the right, showing a total of \$367.93. At the bottom are 'CREATE', 'CANCEL', and 'EQUIVALENT CODE' buttons.

Google Cloud My First Project Search (/) for resources, docs, products, and more Search

← Create an instance CREATE VM FROM... EQUIVALENT CODE

New VM instance
Create a single VM instance from scratch

New VM instance from template
Create a single VM instance from an existing template

New VM instance from machine image
Create a single VM instance from an existing machine image

Marketplace
Deploy a ready-to-go solution onto a VM instance

Requires the Service Account User role (roles/iam.serviceAccountUser) to be set for users who want to access VMs with this service account. [Learn more](#)

Access scopes

- ☒ Allow default access
- ☐ Allow full access to all Cloud APIs
- ☐ Set access for each API

Firewall

Add tags and firewall rules to allow specific network traffic from the Internet

- ☒ Allow HTTP traffic
- ☒ Allow HTTPS traffic
- ☐ Allow Load Balancer Health Checks

Observability - Ops Agent

Monitor your system through collection of logs and key metrics.

- ☐ Install Ops Agent for Monitoring and Logging

Advanced options
Networking, disks, security, management, sole-tenancy

Monthly estimate
\$367.93
That's about \$0.50 hourly
Pay for what you use: no upfront costs and per second billing

Item	Monthly estimate
8 vCPU + 64 GB memory	\$354.93
100 GB balanced persistent disk	\$13.00
Total	\$367.93

[Compute Engine pricing](#)
[LESS](#)

CREATE CANCEL EQUIVALENT CODE

Set up firewall configurations to allow necessary traffic to the VM while blocking unauthorized access.

4 . Creating Compute Engine VM

e . Network Interface

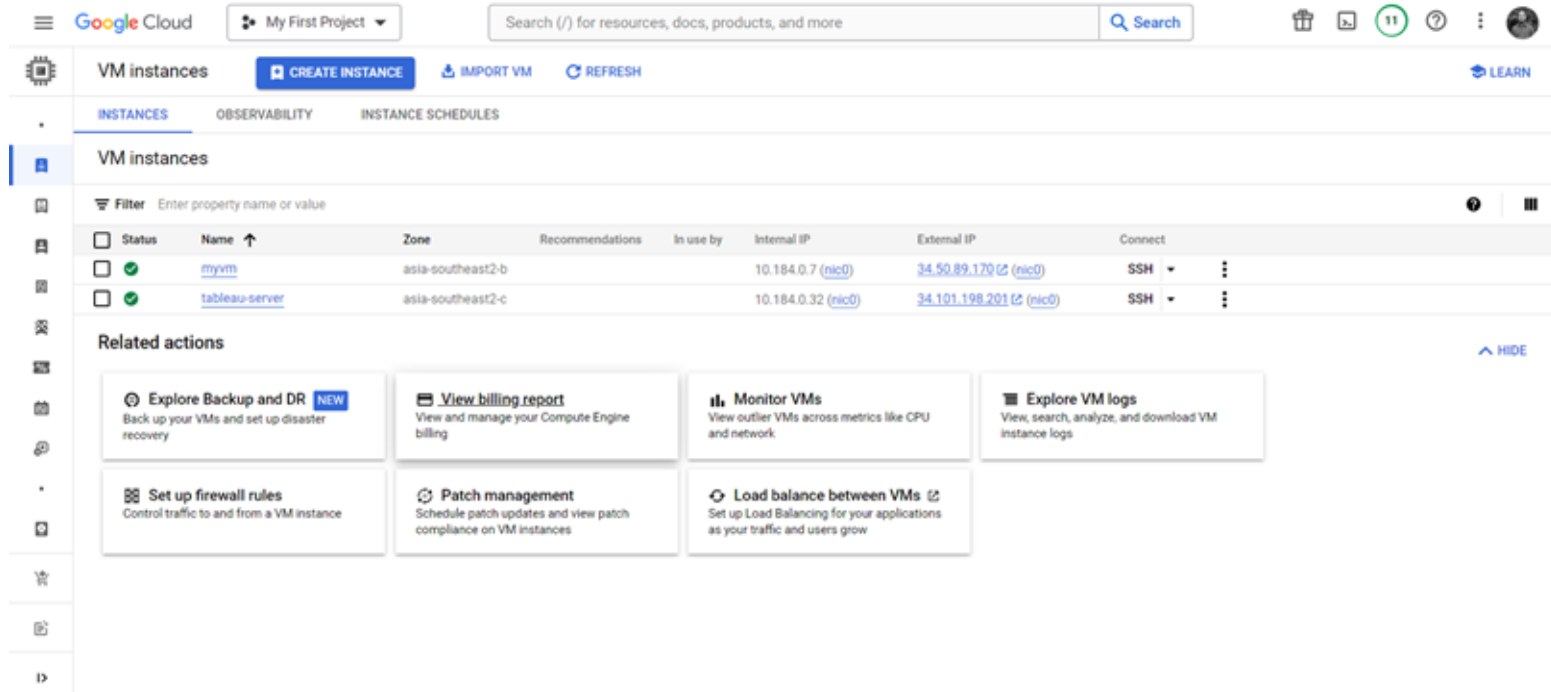
The screenshot shows the 'Create an instance' wizard in the Google Cloud console, specifically the 'Network interfaces' step. The left sidebar lists three options: 'New VM instance' (selected), 'New VM instance from template', and 'New VM instance from machine image'. The main area is titled 'Network interfaces' and includes a sub-header 'New network interface'. It contains several form fields: 'Network' (set to 'tableau-server-network'), 'Subnetwork' (set to 'tableau-server-subnet IPv4 (10.184.0.0/20)'), 'IP stack type' (radio buttons for 'IPv4 (single-stack)' and 'IPv4 and IPv6 (dual-stack)', with 'IPv4 (single-stack)' selected), 'Primary internal IPv4 address' (set to 'Ephemeral (Automatic)'), and 'Alias IP ranges' with an '+ ADD IP RANGE' button. A note states: 'To use IPv6, you need an IPv6 subnet range. [LEARN MORE](#)'. At the bottom are 'CREATE', 'CANCEL', and 'EQUIVALENT CODE' buttons. On the right, a 'Monthly estimate' section shows a total of \$367.93, broken down into '8 vCPU + 64 GB memory' (\$354.93) and '100 GB balanced persistent disk' (\$13.00). A table below this shows the item and its monthly estimate.

Item	Monthly estimate
8 vCPU + 64 GB memory	\$354.93
100 GB balanced persistent disk	\$13.00
Total	\$367.93

Establish the network interface for the VM to facilitate communication with other services and users.

5 . Connect to Compute Engine VM

a . SSH to The VM



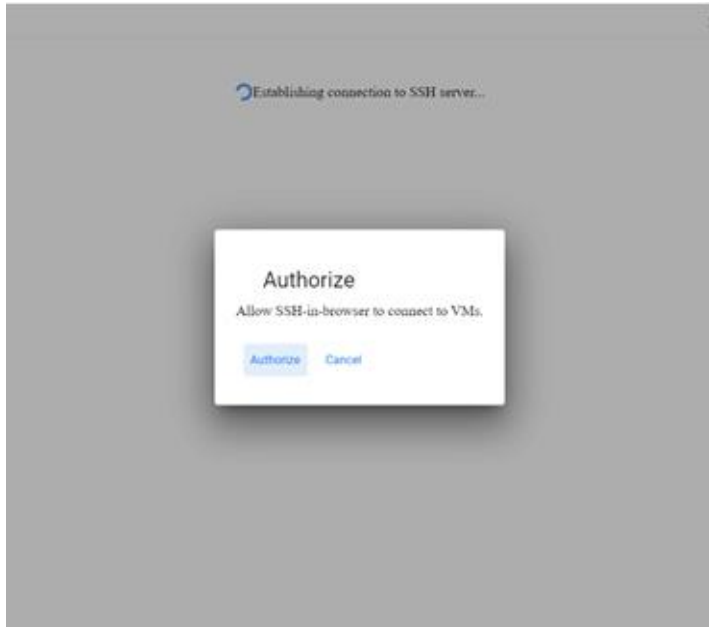
The screenshot shows the Google Cloud console interface for VM instances. At the top, there's a search bar and navigation tabs for 'INSTANCES', 'OBSERVABILITY', and 'INSTANCE SCHEDULES'. Below the tabs, a table lists the VM instances. Two instances are visible: 'mym' and 'tableau-server'. Both are in the 'Running' state (green checkmark) and have an 'SSH' link in the 'Connect' column. The 'mym' instance is in the 'asia-southeast2-b' zone with an internal IP of 10.184.0.7 and an external IP of 34.50.89.170. The 'tableau-server' instance is in the 'asia-southeast2-c' zone with an internal IP of 10.184.0.32 and an external IP of 34.101.198.201. Below the table, there's a 'Related actions' section with several cards: 'Explore Backup and DR', 'View billing report', 'Monitor VMs', 'Explore VM logs', 'Set up firewall rules', 'Patch management', and 'Load balance between VMs'.

Status	Name	Zone	Recommendations	In use by	Internal IP	External IP	Connect
☒	mym	asia-southeast2-b			10.184.0.7 (nic0)	34.50.89.170 (nic0)	SSH
☒	tableau-server	asia-southeast2-c			10.184.0.32 (nic0)	34.101.198.201 (nic0)	SSH

Access the VM via SSH for direct management and configuration.

5 . Connect to Compute Engine VM

b . Authorize SSH



```
SSH-in-browser

Welcome to Ubuntu 22.04.4 LTS (GNU/Linux 6.5.0-1024-gcp x86_64)

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/pro

System information as of Thu Jul 25 12:48:21 UTC 2024

System load:  0.93               Processes:    246
Usage of /:   20.3% of 96.73GB   Users logged in: 0
Memory usage: 35%               IPv4 address for ens4: 10.184.0.32
Swap usage:   0%

* Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
  just raised the bar for easy, resilient and secure K8s cluster deployment.

https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

7 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Thu Jul 25 04:45:28 2024 from 35.235.245.65
ikhsannur1996@tableau-server:~$
```

Ensure SSH connections are authorized for secure access.

6 . Downloading Tableau Server

2024.2	Downloads
2023.3	Downloads
2023.1	Downloads
2022.3	Downloads

Download the Tableau Server installation package using the following command:

```
wget https://downloads.tableau.com/esdalt/2023.1.15/tableau-server-2023-1-15_amd64.deb
```

Linux

- [tableau-tabcmd-2023-1-15.noarch.rpm \(17 MB\)](#)
SHA-1: 7056af6d82bd7bae252579e78374d182cc46b53f
SHA-256: 7f42d539b39c62b9ad76d2e5409ea73f64ce3088eb9225e2945c6fd1704f956
MD5: 884184ef1b1956e17eb36b24b9ef8dd2
- [tableau-tabcmd-2023-1-15_all.deb \(17 MB\)](#)
SHA-1: b6b83f74a1dceaa193af1327685045bfb5f40243
SHA-256: 62cc1d647c9d359402ae8148cf769d928ee9e8c80466235279c5a57e42cee21f
MD5: e90d9965d540a33dad6358bd78e4ad80
- [tableau-server-2023-1-15.x86_64.rpm \(2971 MB\)](#)
SHA-1: 67ceea23ae80aa218f6373680461d59b73820427
SHA-256: 0b66fbcaab41e27dc095b6d5ff90d71a5bc9f2a288c3c643a909a1c5fffb6329
MD5: 52ebb1390fe74cc64cbeb35a7f30cdfc
- [tableau-server-2023-1-15_amd64.deb \(2956 MB\)](#)
SHA-1: 66a7cf6c102990243f63f228bf1d523762f3a1a3
SHA-256: a2389fa48067ec021517ebe7f74406850f5dfeba8997f31e3d9d81c8ab4fc00
MD5: e515cd7546e3403c95386d6790723045
- [tableau-server-container-setup-tool-2023.1.3.tar.gz \(Less than 1 MB\)](#)
SHA-1: ee7e7da3e0c00450b496a658e8d72e200a2a4ee1
SHA-256: 290709a6003f13c6e92cc8df2b5a6b623f0987849433403f3bb8c716b0b203b5
MD5: 3b93e675c2d2d5f7831a8a8d2e0c8905

Refer to the Tableau releases page for the latest download links.
<https://www.tableau.com/support/releases/server>

7 . Installing Tableau Server

Execute the following commands to install Tableau Server:

```
sudo apt-get update
sudo apt-get upgrade
sudo apt-get -y install gdebi-core
sudo gdebi -n tableau-server-<version>_amd64.deb
sudo /opt/tableau/tableau_server/packages/script.tableau-server-<version>/initialize-tsm --
accepteula
```

```
Tableau Server runs best with at least 128 GB memory to run, but found only 63 GB of memory.
See: http://www.tableau.com/products/server/specs

*****
Locale 'C.UTF-8' is unsupported. Using en_US.UTF-8 as a fallback
*****

Creating 'tsmadmin' group for TSM admin authorization
Creating 'tableau' unprivileged user account
Creating environment file...
Creating directories and setting permissions...
Using '/var/opt/tableau/tableau_server' as the data directory.
Adding user 'ikhsannur1996' to group 'tableau'...
Adding user 'ikhsannur1996' to group 'tsmadmin'...
Added. Note: These group membership changes do not take effect in shells already open. For these to take effect
, log out of the shell and log back in.
Starting TSM services...
Updating repository version in Tableau Server Coordination Service.
TSM services started successfully
Use the 'tsm' command to continue setting up Tableau Server.
>> Tableau binary directory will be added to PATH for new shells. To get the
>> updated path, either start a new session, or for bash users run:
>> source /etc/profile.d/tableau_server.sh
The TSM administrative web interface (and REST API) is now available at
https://tableau-server:8850
You can continue the configuration and initialization of Tableau server using either the TSM command line inter
face,
or the web interface.
You will be prompted to authenticate, or can log in using the username 'ikhsannur1996', with the same password
you used to log into this session. You could also use any username, with its password, from the administrative
group in the domain.
Done.
```

8 . Creating TSM User Password

Establish a password for the Tableau Services Manager (TSM) user to secure access.

```
ikhsannur1996@tableau-server:~$ sudo passwd ikhsannur1996
New password:
Retype new password:
passwd: password updated successfully
ikhsannur1996@tableau-server:~$
```

9 . TSM Configuration - Activate

← → ↻ Not secure https://34.101.198.201:8850/#/ ☆ 1

+ tableau

Activate Register Setup Initialize

Enter your license product key to get started with Tableau Server.

Product Key

The key has 20 characters

0000-0000-0000-0000-000

[I can't find my product key.](#)

Tableau Server requires at least one product key that activates the server and specifies the number of license levels you can assign to users. You can access your product keys from the [Tableau Customer Portal](#).

Activated Product Keys
No product key currently activated

Activate Product Key Next

Try it free for 14 days

Start Tableau Server Trial

Questions? Check out the [Administrator Guide](#) or contact [Tableau Customer Service](#)

Activate/Start Trial the TSM service to manage Tableau Server.

10 . TSM Configuration - Register

The screenshot shows a web browser window with the URL `https://34.101.198.201:8850/#/`. The page features the Tableau logo and a progress bar with four steps: **Activate** (completed), **Register** (current step), **Setup**, and **Initialize**. Below the progress bar, a message states: "Register with Tableau. All fields are required." The registration form is divided into two sections: **Contact Information** and **Company Information**.

Contact Information

First Name	Last Name
Mohamad Ikhsan	Nurulloh
Phone Number	Email
085 [redacted]	ikhs [redacted]

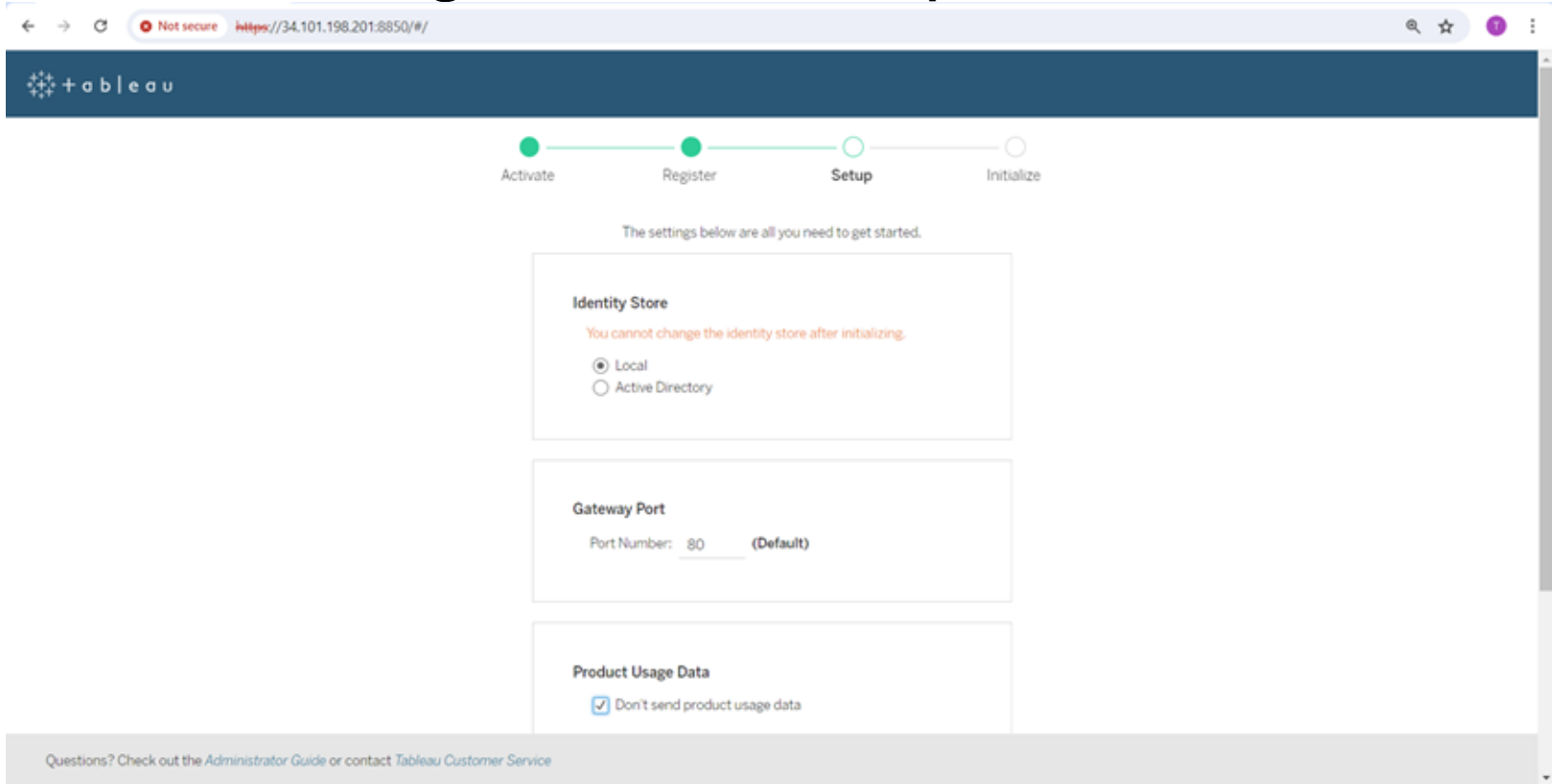
Company Information

Organization	
My Company	
Industry	Company Size

Questions? Check out the [Administrator Guide](#) or contact [Tableau Customer Service](#)

Register the Tableau Server.

11. TSM Configuration - Setup



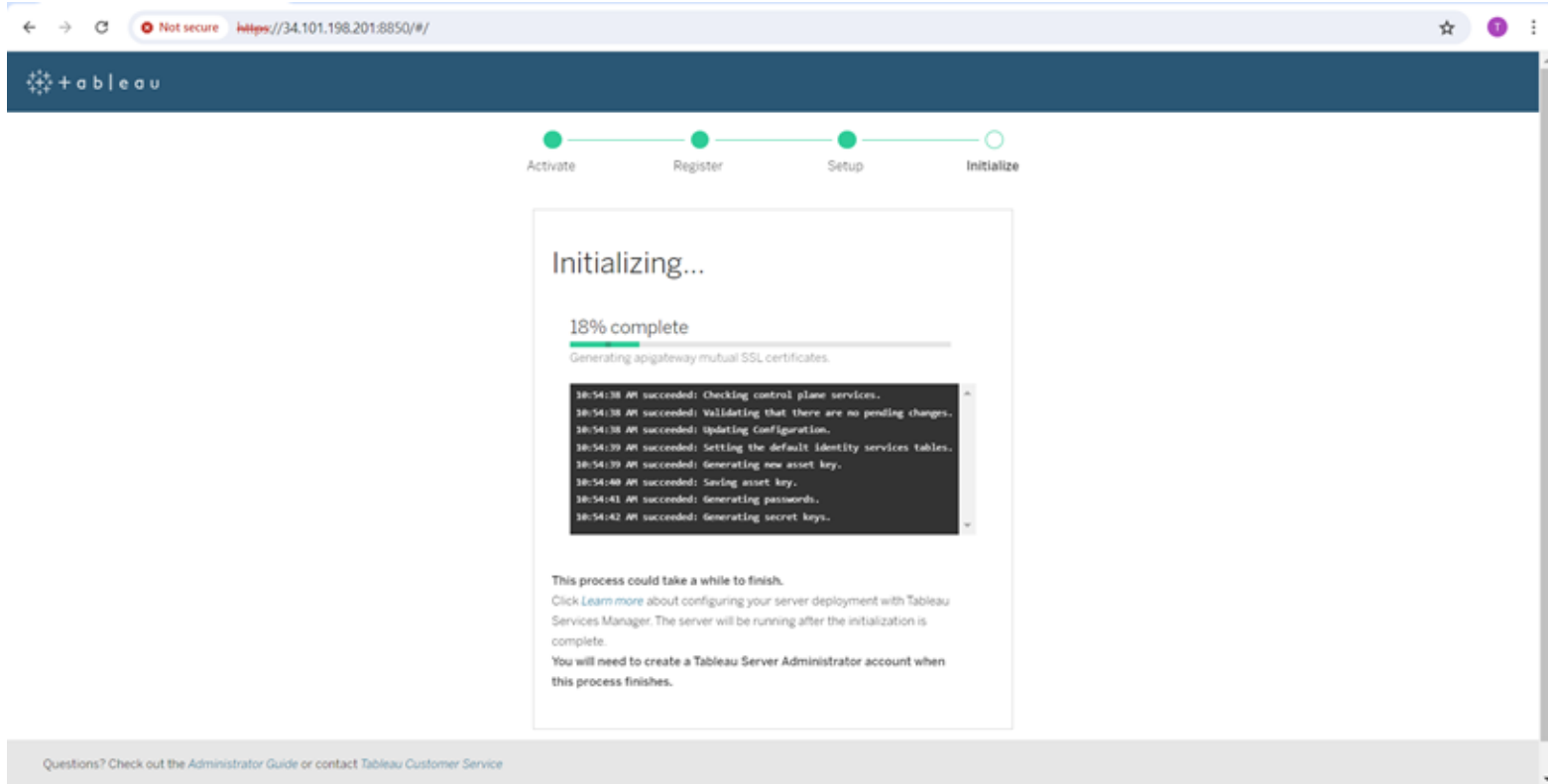
The screenshot shows the Tableau Server Setup page in a web browser. The browser's address bar displays "Not secure" and the URL "https://34.101.198.201:8850/#/". The Tableau logo is in the top left corner. A progress bar at the top indicates the setup stages: Activate (completed), Register (completed), Setup (current stage), and Initialize (pending). Below the progress bar, a message states: "The settings below are all you need to get started." The settings are organized into three sections:

- Identity Store**
You cannot change the identity store after initializing.
☒ Local
☐ Active Directory
- Gateway Port**
Port Number: (Default)
- Product Usage Data**
☒ Don't send product usage data

At the bottom, a footer link reads: "Questions? Check out the [Administrator Guide](#) or contact [Tableau Customer Service](#)".

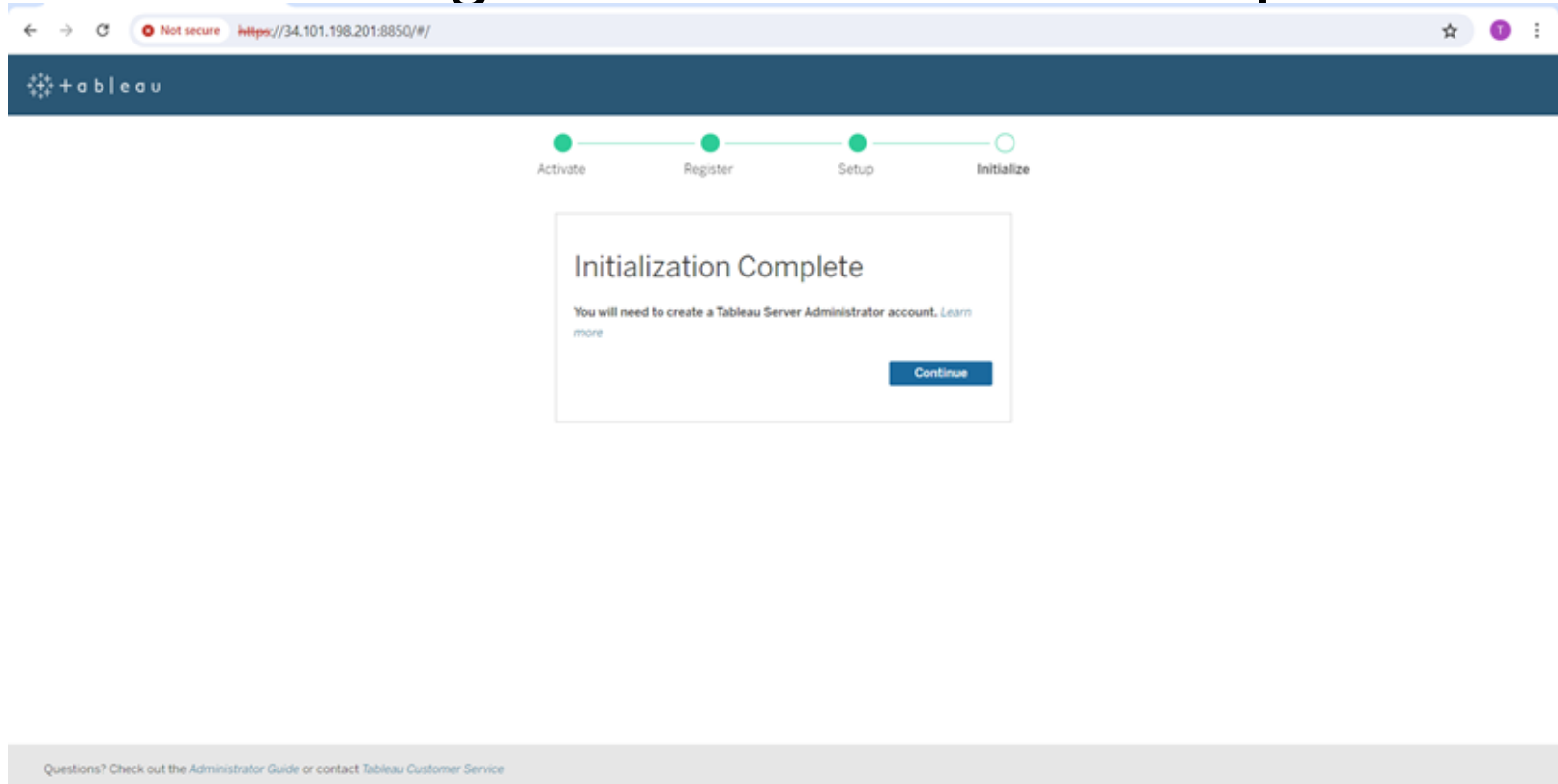
Configure TSM settings according to your organizational needs.

12 . TSM Configuration - Initialize



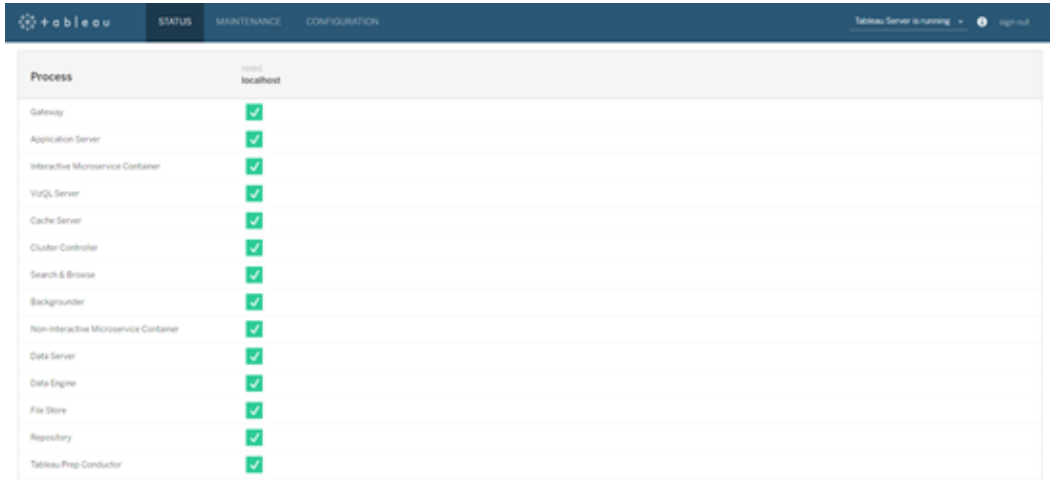
TSM will begin the initialization process.

13 . TSM Configuration - Initialize Complete

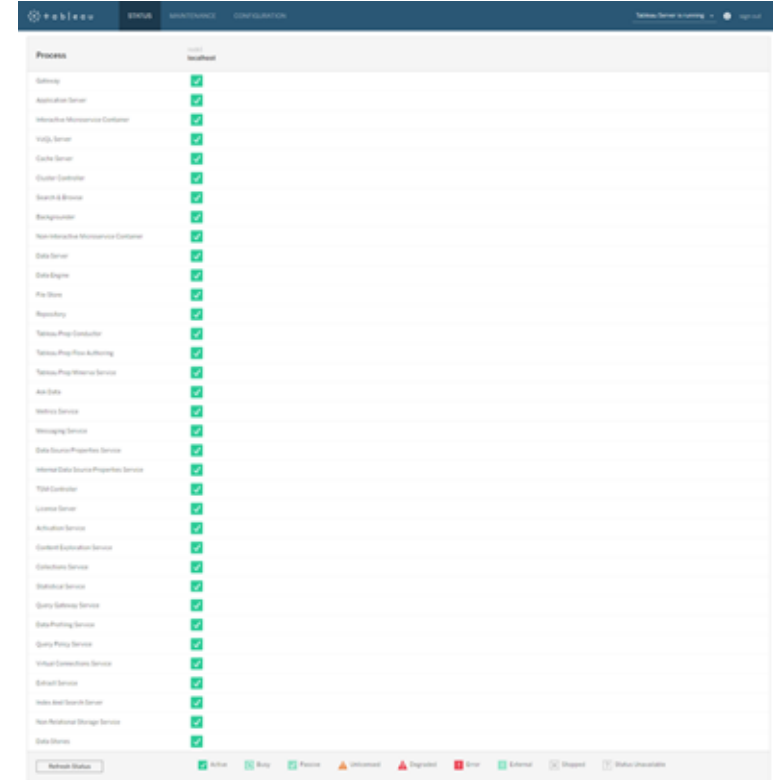


The initialization process for TSM has been completed.

14 . TSM GUI



Process	localhost
Gateway	✓
Application Server	✓
Interactive Microservice Container	✓
ViQL Server	✓
Cache Server	✓
Cluster Controller	✓
Search & Browse	✓
Background	✓
Non-Interactive Microservice Container	✓
Data Server	✓
Data Engine	✓
File Store	✓
Repository	✓
Tableau Prep Conductor	✓



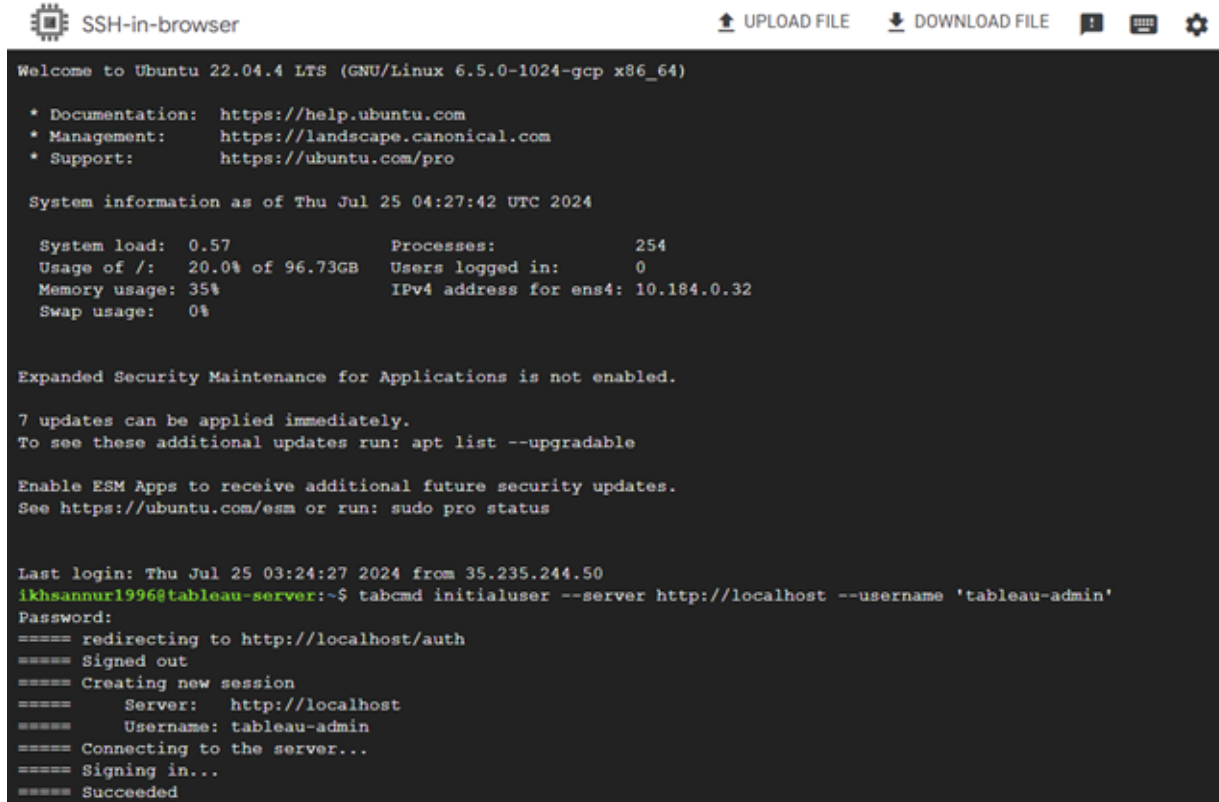
Process	localhost
Gateway	✓
Application Server	✓
Interactive Microservice Container	✓
ViQL Server	✓
Cache Server	✓
Cluster Controller	✓
Search & Browse	✓
Background	✓
Non-Interactive Microservice Container	✓
Data Server	✓
Data Engine	✓
File Store	✓
Repository	✓
Tableau Prep Conductor	✓
Tableau Prep Flow Authoring	✓
Tableau Prep Warehouse Service	✓
Alert Data	✓
Webhooks Service	✓
Messaging Service	✓
Data Source Properties Service	✓
Internal Data Source Properties Service	✓
TSM Conductor	✓
Licensed Server	✓
Activation Service	✓
Content Evaluation Service	✓
Content Service	✓
Database Service	✓
Query Gateway Service	✓
Query Profiling Service	✓
Query Policy Service	✓
Virtual Connections Service	✓
Extract Service	✓
Index And Search Server	✓
Non-Replicated Storage Service	✓
Data Stores	✓

Refresh Status

Active Busy Paused Unresponsive Ungranted Error External Disabled Unavailable

Open the TSM graphical user interface for management.

15 . TSM Configuration - Initial Tableau Server User



```
SSH-in-browser  UPLOAD FILE  DOWNLOAD FILE  !  [terminal icon]  [gear icon]

Welcome to Ubuntu 22.04.4 LTS (GNU/Linux 6.5.0-1024-gcp x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu Jul 25 04:27:42 UTC 2024

System load:  0.57               Processes:            254
Usage of /:   20.0% of 96.73GB   Users logged in:     0
Memory usage: 35%               IPv4 address for ens4: 10.184.0.32
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

7 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Thu Jul 25 03:24:27 2024 from 35.235.244.50
ikhsannur1996@tableau-server:~$ tabcmd initialuser --server http://localhost --username 'tableau-admin'
Password:
===== redirecting to http://localhost/auth
===== Signed out
===== Creating new session
=====   Server:   http://localhost
=====   Username: tableau-admin
===== Connecting to the server...
===== Signing in...
===== Succeeded
```

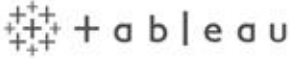
Set up the initial user for Tableau Server login.

16 . Login to Tableau Server Using Initial User



The screenshot shows a web browser window with the address bar displaying "Not secure 34.101.198.201/#/signin". The main content area features the Tableau logo, which consists of a grid of plus signs followed by the word "tableau". Below the logo, there are two input fields: "Username" with the text "tableau-admin" and "Password" with masked characters "*****". A blue "Sign In" button is positioned below the password field.

← → ↻ ⚠ Not secure 34.101.198.201/#/signin

 + a b l e a u

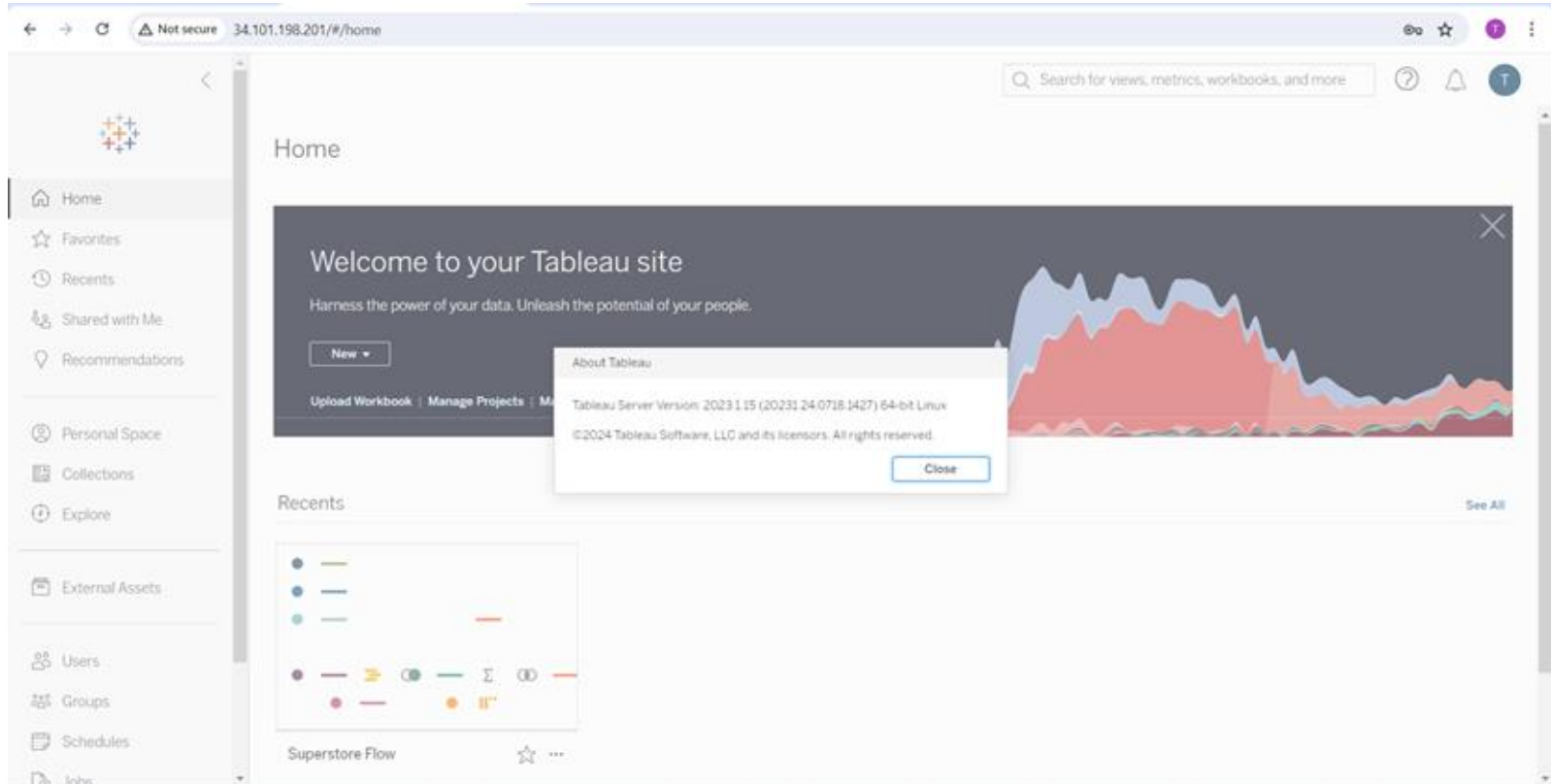
Username
tableau-admin

Password

Sign In

Use the initial user credentials to access Tableau Server.

17 . Ensure Installation Was Successful



Verify that Tableau Server has been installed successfully.

Driver Installation

1 . Installing Data Connectors Driver

Download and install necessary data connector drivers for Tableau from the official Tableau website .



PostgreSQL

Tableau Server: 2021.1 - 2024.2

Complete the following steps to install the PostgreSQL driver.

1. Download the Java 8 JDBC driver from <https://jdbc.postgresql.org/download/>
2. Copy the .jar file to the following folder (you may have to create it manually).

```
/opt/tableau/tableau_driver/jdbc
```

Resources

- [Database Drivers](#)

Tableau Bridge:

Complete the following steps to install the PostgreSQL driver.

1. Download the Java 8 JDBC driver from <https://jdbc.postgresql.org/download/>
2. Copy the .jar file to the following folder (you may have to create it manually).

```
/opt/tableau/tableau_driver/jdbc
```

<https://www.tableau.com/support/drivers>

2 . Restarting TSM

The screenshot shows the Tableau Server Administration console interface. The top navigation bar includes 'STATUS', 'MAINTENANCE', and 'CONFIGURATION'. The left sidebar lists 'Topology', 'Security', 'User Identity & Access', 'Notifications', and 'Licensing'. The main content area is titled 'User Identity & Access' and contains tabs for 'Identity Store', 'Authentication Method', 'Trusted Authentication', and 'Authorization Server'. The 'Trusted Authentication' tab is active, showing configuration options for 'Trusted hosts' (103.23.198.x) and 'Token Length' (24). A modal dialog titled 'Applying Changes in Progress' is displayed in the foreground, indicating that 77% of the changes are complete. The dialog includes a progress bar and a log of system messages.

Applying Changes in Progress

77% complete

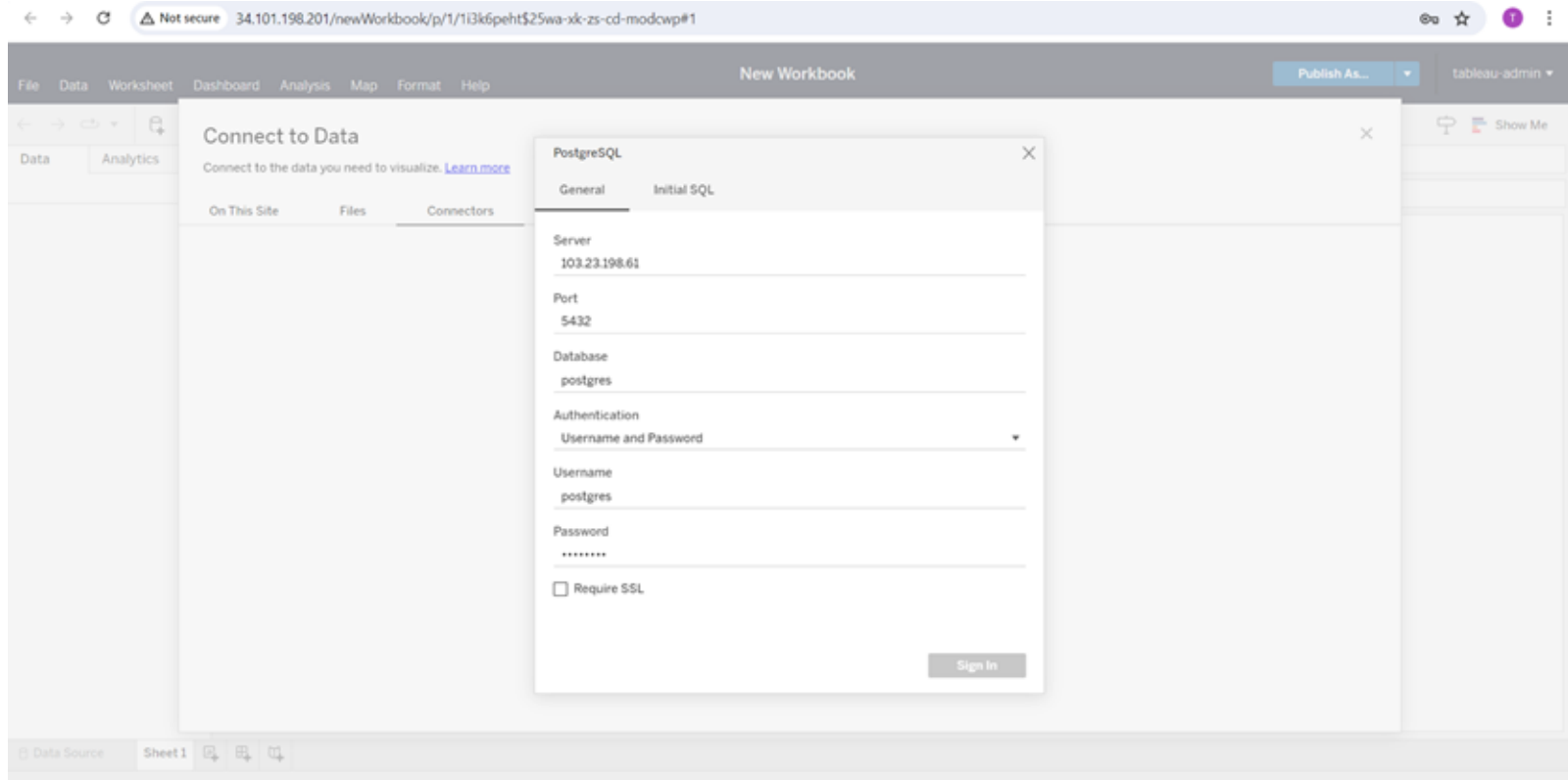
Waiting for services to reconfigure.

```
12:04:25 PM succeeded: Applying the configuration to deploy.  
12:04:26 PM succeeded: Validating the new topology.  
12:04:26 PM succeeded: Determining if server needs to be started.  
12:04:26 PM succeeded: Disabling all services.  
12:04:27 PM succeeded: Waiting for the services to stop.  
12:04:28 PM succeeded: Updating nodes to new topology.  
12:04:28 PM succeeded: Waiting for topology to be applied.  
12:04:28 PM succeeded: Updating nodes to new configuration.  
12:04:29 PM succeeded: Reconfiguring services.
```

We don't have information about how long this job runs.
[Learn more about running jobs with Tableau Services Manager.](#)

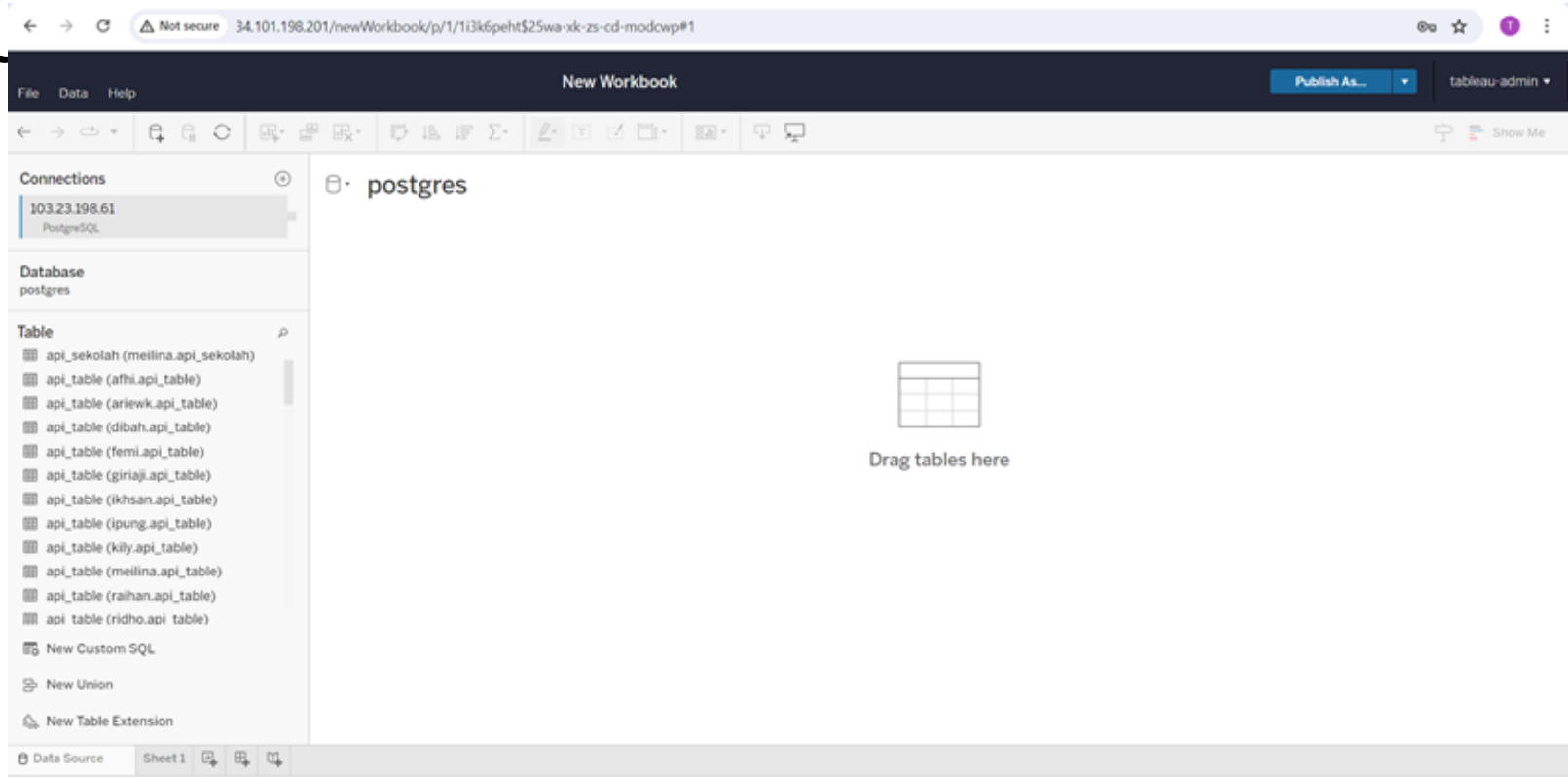
Restart TSM to apply changes after driver installation.

3 . Data Connectors Driver Testing



Ensure that the installed data connectors function correctly by performing tests.

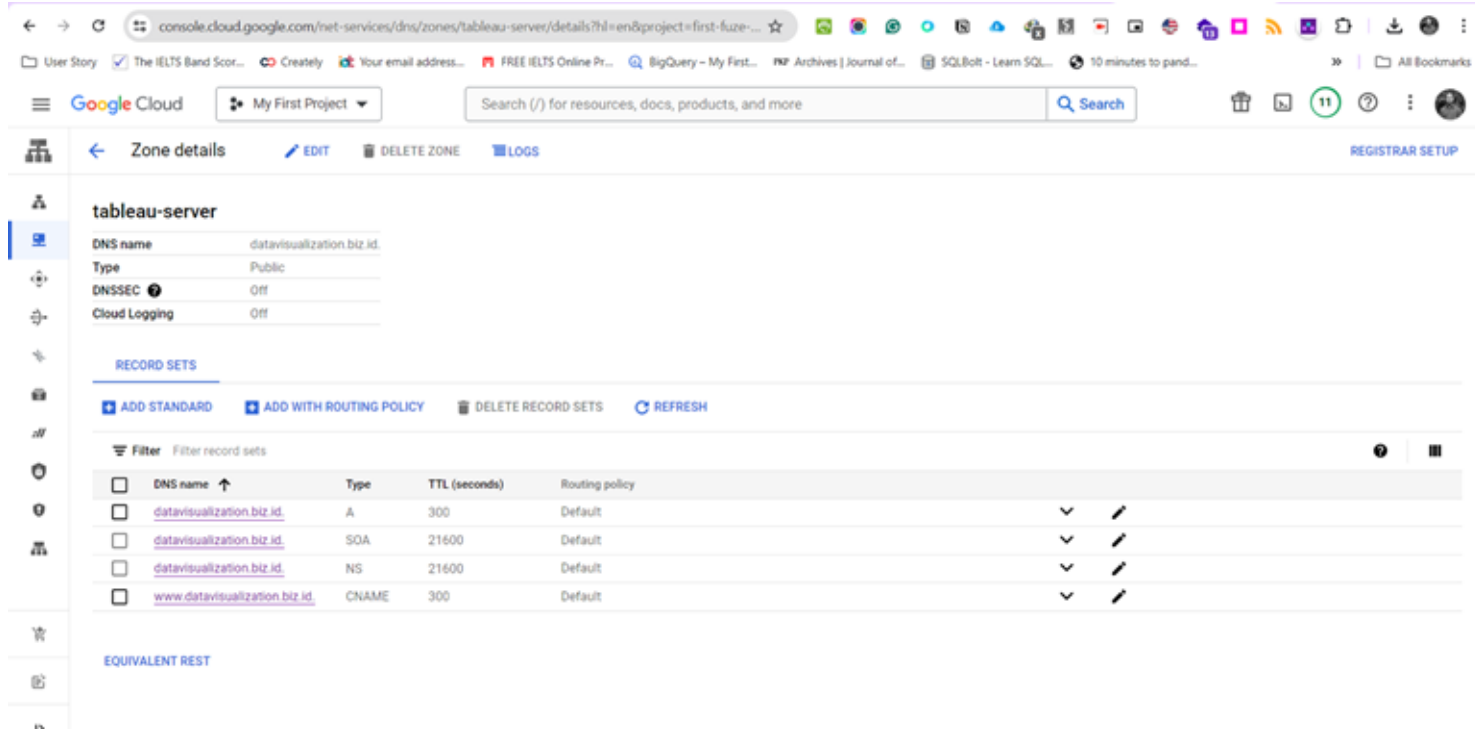
4 . Ensuing The Data Connection Was Successful



Data Connectors Driver Testing Success

DNS Configuration

1 . Configuring Cloud DNS



The screenshot shows the Google Cloud DNS console interface. The browser address bar displays the URL: `console.cloud.google.com/net-services/dns/zones/tableau-server/details?hl=en&project=first-fuze...`. The page title is "Zone details" for the "tableau-server" zone. The left sidebar shows the "tableau-server" zone selected. The main content area displays the zone details and a list of record sets.

Zone details:

- DNS name: `datavisualization.biz.id`
- Type: Public
- DNSSEC: Off
- Cloud Logging: Off

RECORD SETS

Buttons: ADD STANDARD, ADD WITH ROUTING POLICY, DELETE RECORD SETS, REFRESH

Filter: Filter record sets

☐	DNS name ↑	Type	TTL (seconds)	Routing policy		
☐	datavisualization.biz.id	A	300	Default	▼	✎
☐	datavisualization.biz.id	SOA	21600	Default	▼	✎
☐	datavisualization.biz.id	NS	21600	Default	▼	✎
☐	www.datavisualization.biz.id	CNAME	300	Default	▼	✎

EQUIVALENT REST

Set up Cloud DNS for domain name management, following the guidelines provided by Google .
<https://cloud.google.com/dns/docs/set-up-dns-records-domain-name>
<https://www.youtube.com/watch?v=g1KKI7JVJH8>

2 . Configuring Cloud DNS

Google Cloud | My First Project | Search () for resources, docs, products, and more

Resource record set details | EDIT RESOURCE RECORD SET | DELETE RESOURCE RECORD SET

DNS name: datausualization.biz id
Type: NS
TTL(seconds): 21400

Routing data

Filter: Filter routing data

Data

ns-cloud-e1.gcpelddomains.com.
ns-cloud-e2.gcpelddomains.com.
ns-cloud-e3.gcpelddomains.com.
ns-cloud-e4.gcpelddomains.com.

EQUIVALENT REST

Google Cloud | My First Project | Search () for resources, docs, products, and more

Resource record set details | EDIT RESOURCE RECORD SET | CHANGE ROUTING POLICY | DELETE RESOURCE RECORD SET

DNS name: www.datausualization.biz id
Type: CHAOS
TTL(seconds): 300

Routing data

Filter: Filter routing data

Data

datausualization.biz id

EQUIVALENT REST

Google Cloud | My First Project | Search () for resources, docs, products, and more

Resource record set details | EDIT RESOURCE RECORD SET | DELETE RESOURCE RECORD SET

DNS name: datausualization.biz id
Type: SOA
TTL(seconds): 21400

Routing data

Filter: Filter routing data

Data

ns-cloud-e1.gcpelddomains.com. cloud-dns-hostmaster.google.com. 1 21400 3600 259200 300

EQUIVALENT REST

Google Cloud | My First Project | Search () for resources, docs, products, and more

Resource record set details | EDIT RESOURCE RECORD SET | CHANGE ROUTING POLICY | DELETE RESOURCE RECORD SET

DNS name: datausualization.biz id
Type: A
TTL(seconds): 300

Routing data

Filter: Filter routing data

Data

34.101.198.201

EQUIVALENT REST

3. Configuring Name Server on Your DNS Provider

The screenshot shows the DomaiNesia web interface for managing the domain `datavisualization.biz.id`. The left sidebar contains a 'Main Menu' with links to Dashboard, My Services, Domains (highlighted), Billing, and Support, along with a 'Benefits Menu' for Affiliates and Promotion. The main content area is titled 'Managing datavisualization.biz.id' and includes a 'Nameservers' section. This section displays the current nameservers: `ns-cloud-e1.googledomains.com`, `ns-cloud-e2.googledomains.com`, `ns-cloud-e3.googledomains.com`, and `ns-cloud-e4.googledomains.com`. Below this, there are radio buttons to select between 'Use default nameservers', 'DomaiNesia hosting nameservers', and 'Use custom nameservers (enter below)'. The 'Use custom nameservers' option is selected, and the current nameservers are listed in input fields below.

DomaiNesia

Managing datavisualization.biz.id

Dashboard / MyDomaiNesia / My Domains / datavisualization.biz.id

Mohamad Ikhsan Nurulloh

Main Menu

- Dashboard
- My Services
- Domains
- Billing
- Support

Benefits Menu

- Affiliates
- Promotion

Order

Hak Cipta © 2024 PT Deneva

Manage

- Overview
- Nameservers
- Registrar Lock
- Addons
- Contact Information
- Private Nameservers
- DNS Management
- DNSSEC Management
- Domain Forwarding
- Blogger Settings
- Get EPP Code

Actions

- Renew
- Register a New Domain

Nameservers

You can change where your domain points to here. Please be aware changes can take up to 24 hours to propagate.

Your Current Nameservers:

- Nameserver 1: ns-cloud-e1.googledomains.com
- Nameserver 2: ns-cloud-e2.googledomains.com
- Nameserver 3: ns-cloud-e3.googledomains.com
- Nameserver 4: ns-cloud-e4.googledomains.com

☐ Use default nameservers

☐ DomaiNesia hosting nameservers

☒ Use custom nameservers (enter below)

Nameserver 1: ns-cloud-e1.googledomains.com

Nameserver 2: ns-cloud-e2.googledomains.com

Nameserver 3: ns-cloud-e3.googledomains.com

Nameserver 4: ns-cloud-e4.googledomains.com

Configure the name server with your DNS provider to ensure proper domain resolution. Use services like DomaiNesia for DNS management. Get DNS and SSL: <https://www.domainesia.com/>

4 . Ping Testing

```
Windows PowerShell
PS C:\Users\user> ping 34.101.198.201

Pinging 34.101.198.201 with 32 bytes of data:
Reply from 34.101.198.201: bytes=32 time=24ms TTL=57
Reply from 34.101.198.201: bytes=32 time=78ms TTL=57
Reply from 34.101.198.201: bytes=32 time=282ms TTL=57
Reply from 34.101.198.201: bytes=32 time=123ms TTL=57

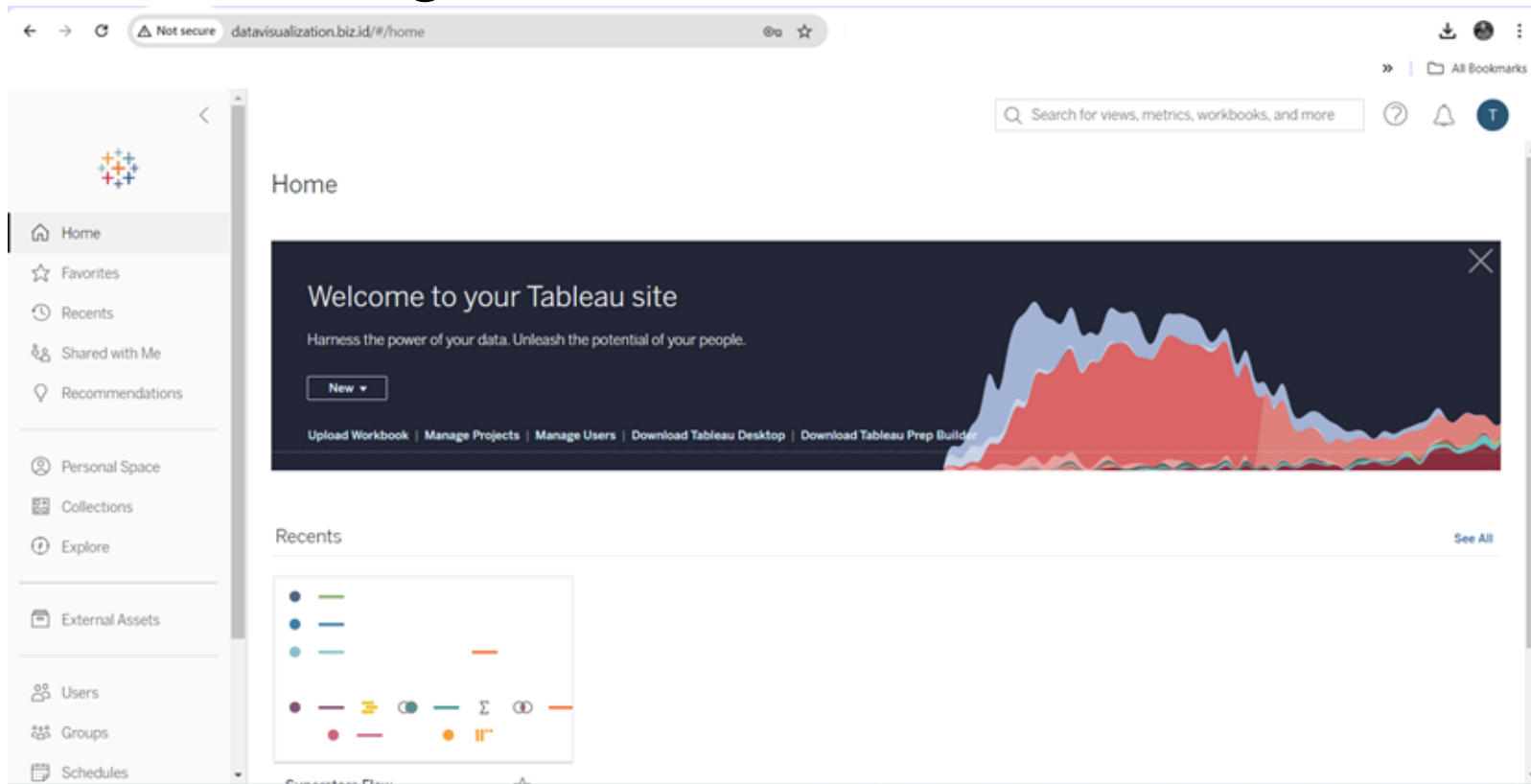
Ping statistics for 34.101.198.201:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 24ms, Maximum = 282ms, Average = 126ms
PS C:\Users\user> ping datavisualization.biz.id

Pinging datavisualization.biz.id [34.101.198.201] with 32 bytes of data:
Reply from 34.101.198.201: bytes=32 time=68ms TTL=57
Reply from 34.101.198.201: bytes=32 time=43ms TTL=57
Reply from 34.101.198.201: bytes=32 time=28ms TTL=57
Reply from 34.101.198.201: bytes=32 time=38ms TTL=57

Ping statistics for 34.101.198.201:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 28ms, Maximum = 68ms, Average = 44ms
```

Conduct ping tests to verify DNS configuration and connectivity.

5 . DNS Change Success



Ensure that DNS changes have been successfully implemented.

SSL Configuration

1 . Upload The Certificate and Key on TSM

The screenshot shows the Tableau Server Configuration interface. The browser address bar indicates the URL is `https://34.101.198.201:8850/#/configuration/security/external-ssl`. The Tableau logo and navigation tabs (STATUS, MAINTENANCE, CONFIGURATION) are at the top. The left sidebar lists navigation options: Topology, Security (selected), User Identity & Access, Notifications, and Licensing. The main content area is titled 'Security' with a subtitle 'Configure Run As service account, enable and configure external SSL, and enable and configure repository (internal) SSL'. There are two tabs: 'External SSL' (active) and 'Repository SSL'. Under 'External web server SSL', there is a link to 'Learn more' and a checked checkbox for 'Enable SSL for server communication'. Below this are four fields for SSL configuration: 'SSL certificate file (Required)' with value 'datavisualization_biz_id.crt' and a 'Select File' button; 'SSL certificate key file (Required)' with value 'datavisualization_biz_id.key' and a 'Select File' button; 'SSL certificate key passphrase' with a text input containing 'Optional'; and 'SSL certificate chain file' with a 'Select File' button. A note at the bottom states: 'After you have configured external SSL, you can enable and configure mutual SSL authentication on Tableau Server. Click here to configure mutual SSL.' At the bottom of the form are 'Cancel' and 'Save Pending Changes' buttons.

← → ↻ Not secure <https://34.101.198.201:8850/#/configuration/security/external-ssl> ☆ ⓘ ⋮

tableau STATUS MAINTENANCE **CONFIGURATION** Tableau Server is running ⓘ sign out

Topology

Security

User Identity & Access

Notifications

Licensing

Security
Configure Run As service account, enable and configure external SSL, and enable and configure repository (internal) SSL.

External SSL Repository SSL

External web server SSL
Configure SSL for encrypted communication between Tableau Server and clients. [Learn more](#)

☒ Enable SSL for server communication

SSL certificate file (Required) **datavisualization_biz_id.crt**

SSL certificate key file (Required) **datavisualization_biz_id.key**

SSL certificate key passphrase

SSL certificate chain file

After you have configured external SSL, you can enable and configure mutual SSL authentication on Tableau Server. [Click here to configure mutual SSL.](#)

Securely upload the SSL certificate and key to TSM to enable HTTPS.

2. Restart TSM

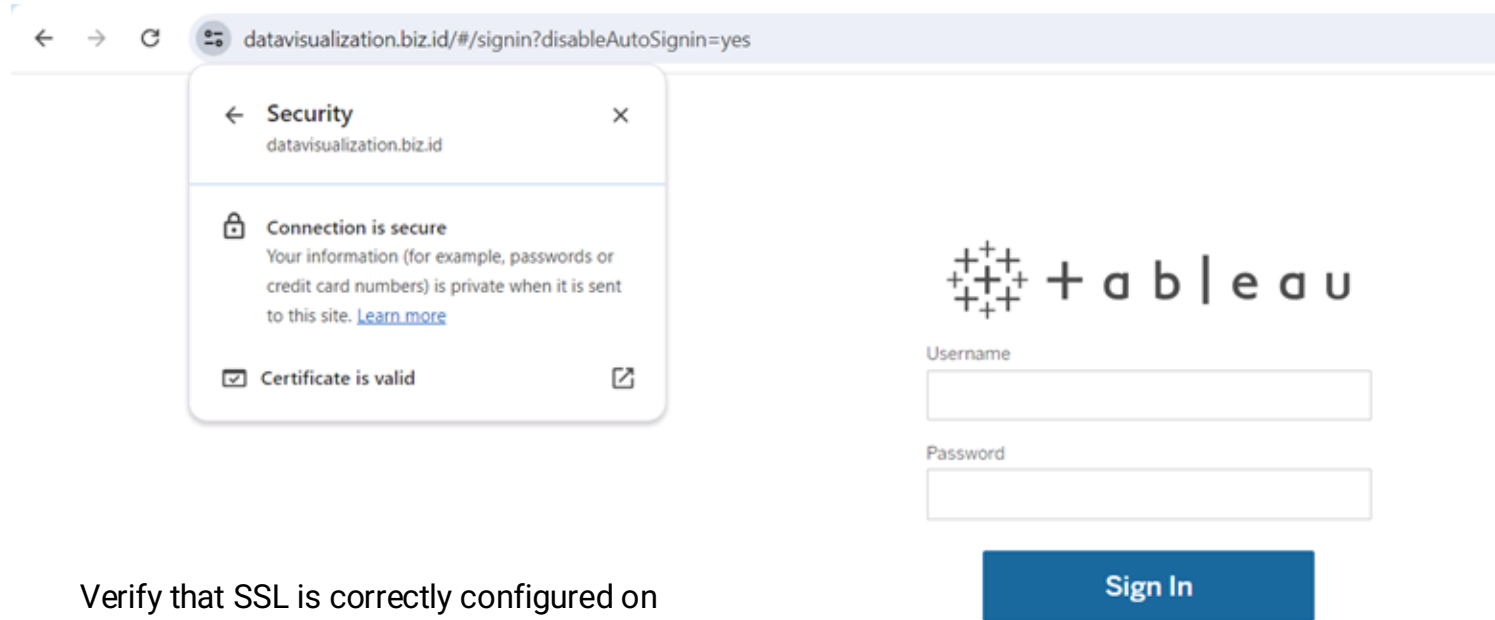
The screenshot shows the Tableau Server Configuration interface. The left sidebar contains navigation links: Topology, Security, User Identity & Access, Notifications, and Licensing. The main content area is titled 'Security' and includes a sub-section for 'External SSL'. The 'External web server SSL' section is active, showing options to 'Enable SSL for server communication' (checked) and fields for 'SSL certificate file', 'SSL certificate key file', 'SSL certificate key passphrase', and 'SSL certificate chain file'. A modal dialog titled 'Applying Changes in Progress' is overlaid on the configuration page. The dialog shows a progress bar at 77% complete and a log of successful configuration steps. The log entries are as follows:

- 2:04:39 PM succeeded: Retrieving the configuration to deploy.
- 2:04:39 PM succeeded: Validating the new topology.
- 2:04:39 PM succeeded: Determining if server needs to be started.
- 2:04:40 PM succeeded: Disabling all services.
- 2:05:55 PM succeeded: Waiting for the services to stop.
- 2:05:55 PM succeeded: Updating nodes to new topology.
- 2:05:56 PM succeeded: Waiting for topology to be applied.
- 2:05:56 PM succeeded: Updating nodes to new configuration.
- 2:05:56 PM succeeded: Reconfiguring services.

The dialog also indicates that the last successful run of this job took 21 minute(s) and provides a link to learn more about running jobs with Tableau Services Manager.

Restart TSM to apply SSL configurations.

3 . Check SSL on Tableau Server



The screenshot shows a web browser window with the address bar displaying `datavisualization.biz.id/#/signin?disableAutoSignIn=yes`. A security warning dialog box is open, titled "Security" with a close button (X). The dialog indicates that the connection is secure and the certificate is valid. The main content area shows the Tableau logo and a sign-in form with fields for "Username" and "Password", and a "Sign In" button.

← → ↻ datavisualization.biz.id/#/signin?disableAutoSignIn=yes

← Security datavisualization.biz.id X

🔒 Connection is secure
Your information (for example, passwords or credit card numbers) is private when it is sent to this site. [Learn more](#)

✅ Certificate is valid ↗

 + a b | e a u

Username

Password

Sign In

Verify that SSL is correctly configured on the Tableau Server.

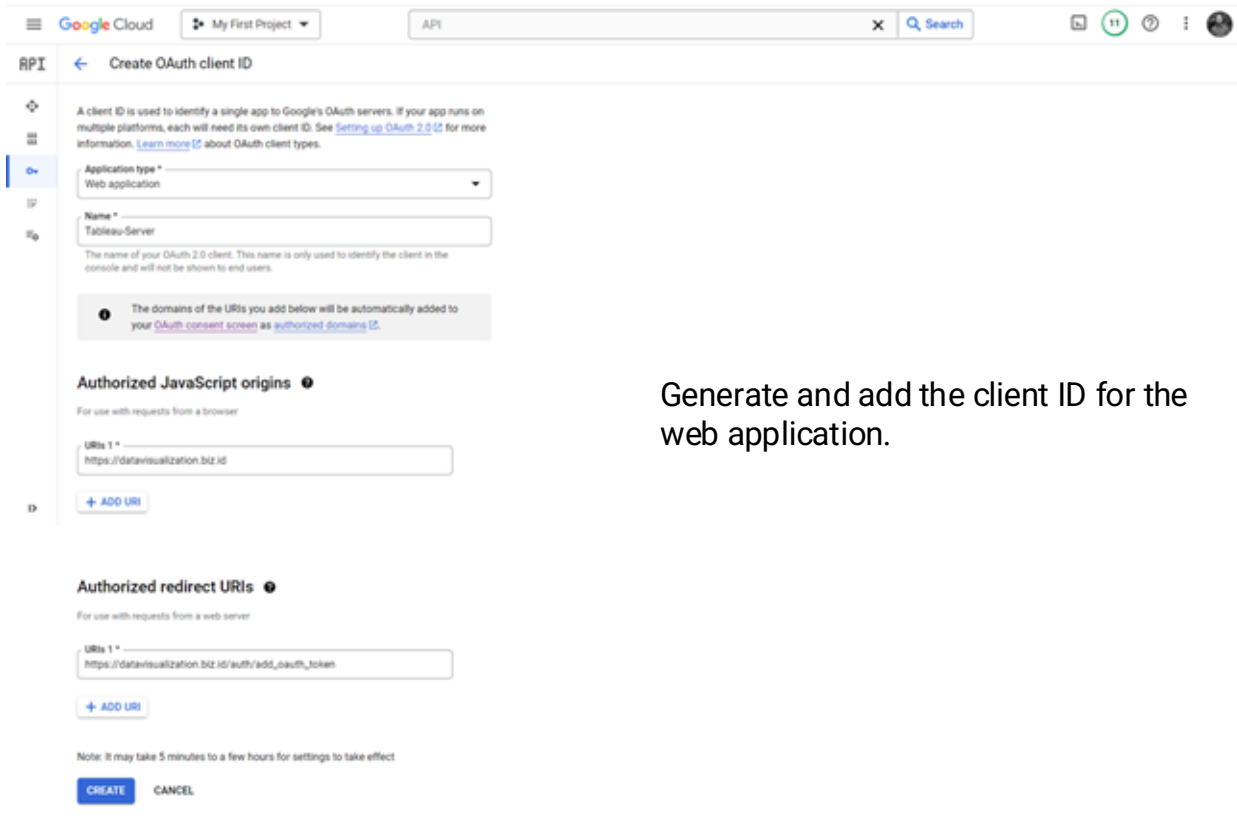
Connect to BigQuery

1 . Add User to OAuth consent screen

The screenshot shows the Google Cloud console interface for configuring an OAuth consent screen. At the top, the Google Cloud logo and 'My First Project' are visible. A search bar is present with the text 'Search (/) for resources, docs, products, and more'. The main heading is 'OAuth consent screen'. Below this, there is a '+ ADD USERS' button. A 'Filter' section allows entering a property name or value. Under 'User information', the email 'ikhsannur1996@gmail.com' is listed with a delete icon. A 'SHOW LESS' link is available. The 'OAuth rate limits' section explains the token grant rate limit, stating the current limit is 10,000 grants per day. It includes a toggle for the time frame, currently set to '1 day' (checked) over '5 minutes'. A line chart shows the rate limit over time, with a red dashed line indicating 'No data is available for the selected time frame' and a red marker at 10,000. The x-axis of the chart shows times from 12 PM to 9 AM on Mon 29.

Configure user access for OAuth authentication with Google services.

2 . Add Credential Client ID for Web application



The screenshot shows the 'Create OAuth client ID' page in the Google Cloud console. The page is for 'My First Project' and is under the 'API' section. The 'Application type' is set to 'Web application'. The 'Name' is 'Tableau-Server'. There is a note about domains being added to the OAuth consent screen. Under 'Authorized JavaScript origins', there is one URI: 'https://datavisualization.biz.id'. Under 'Authorized redirect URIs', there is one URI: 'https://datavisualization.biz.id/auth/add_oauth_token'. At the bottom, there are 'CREATE' and 'CANCEL' buttons.

Google Cloud My First Project API Search

← Create OAuth client ID

A client ID is used to identify a single app to Google's OAuth servers. If your app runs on multiple platforms, each will need its own client ID. See [Setting up OAuth 2.0](#) for more information. [Learn more](#) about OAuth client types.

Application type *
Web application

Name *
Tableau-Server

The name of your OAuth 2.0 client. This name is only used to identify the client in the console and will not be shown to end users.

The domains of the URIs you add below will be automatically added to your [OAuth consent screen](#) as [authorized domains](#).

Authorized JavaScript origins ⓘ
For use with requests from a browser

URIs 1 *
https://datavisualization.biz.id

+ ADD URI

Authorized redirect URIs ⓘ
For use with requests from a web server

URIs 1 *
https://datavisualization.biz.id/auth/add_oauth_token

+ ADD URI

Note: It may take 5 minutes to a few hours for settings to take effect

CREATE CANCEL

Generate and add the client ID for the web application.

3 . Get Client ID and Client Secret

Additional information

Client ID	399288505099-u3t801086h6juiv1sf4khvbsd7cvugr.apps.googleusercontent.com
Creation date	July 26, 2024 at 10:46:04 AM GMT+7

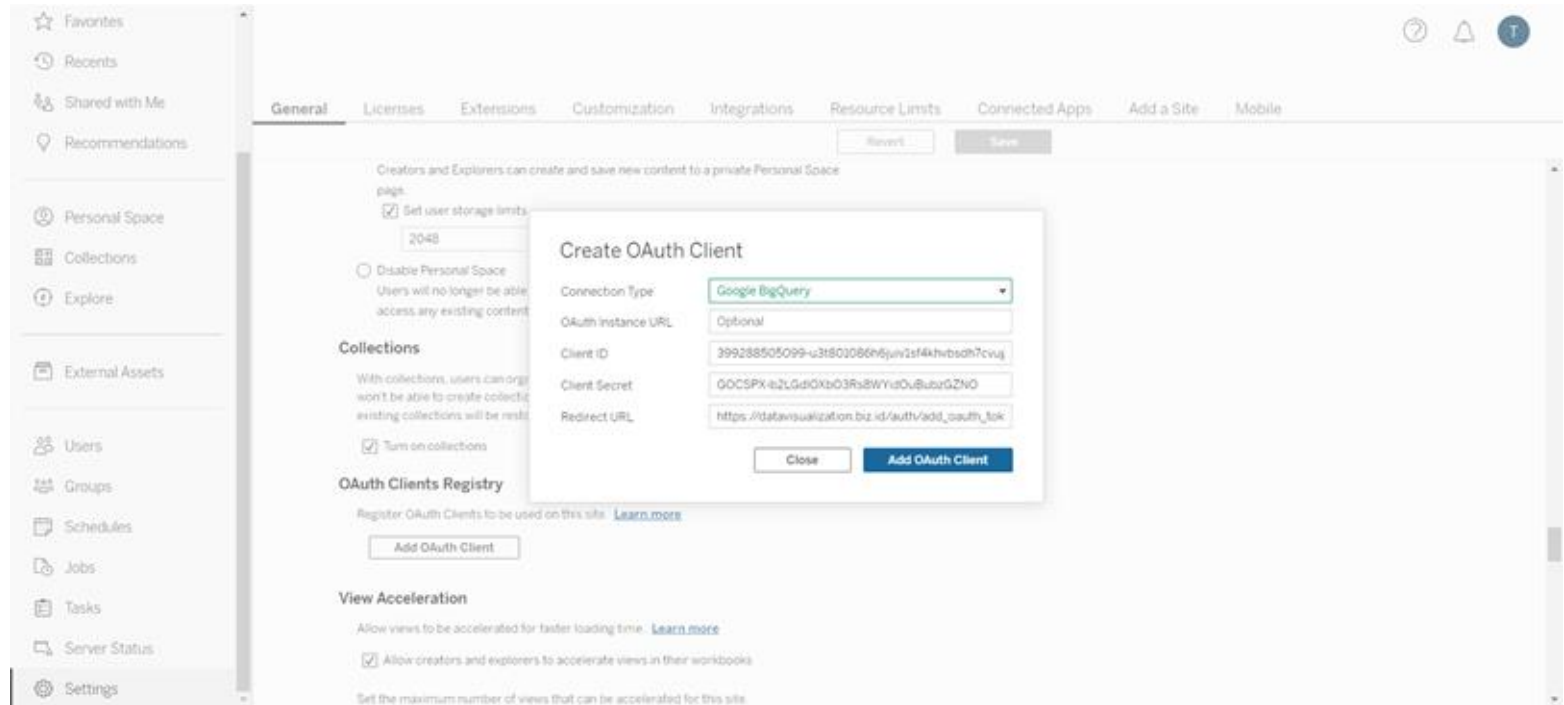
Client secrets

If you are in the process of changing client secrets, you can manually rotate them without downtime. [Learn more](#) 

Client secret	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	 
Creation date	July 26, 2024 at 10:46:04 AM GMT+7	
Status	 Enabled	

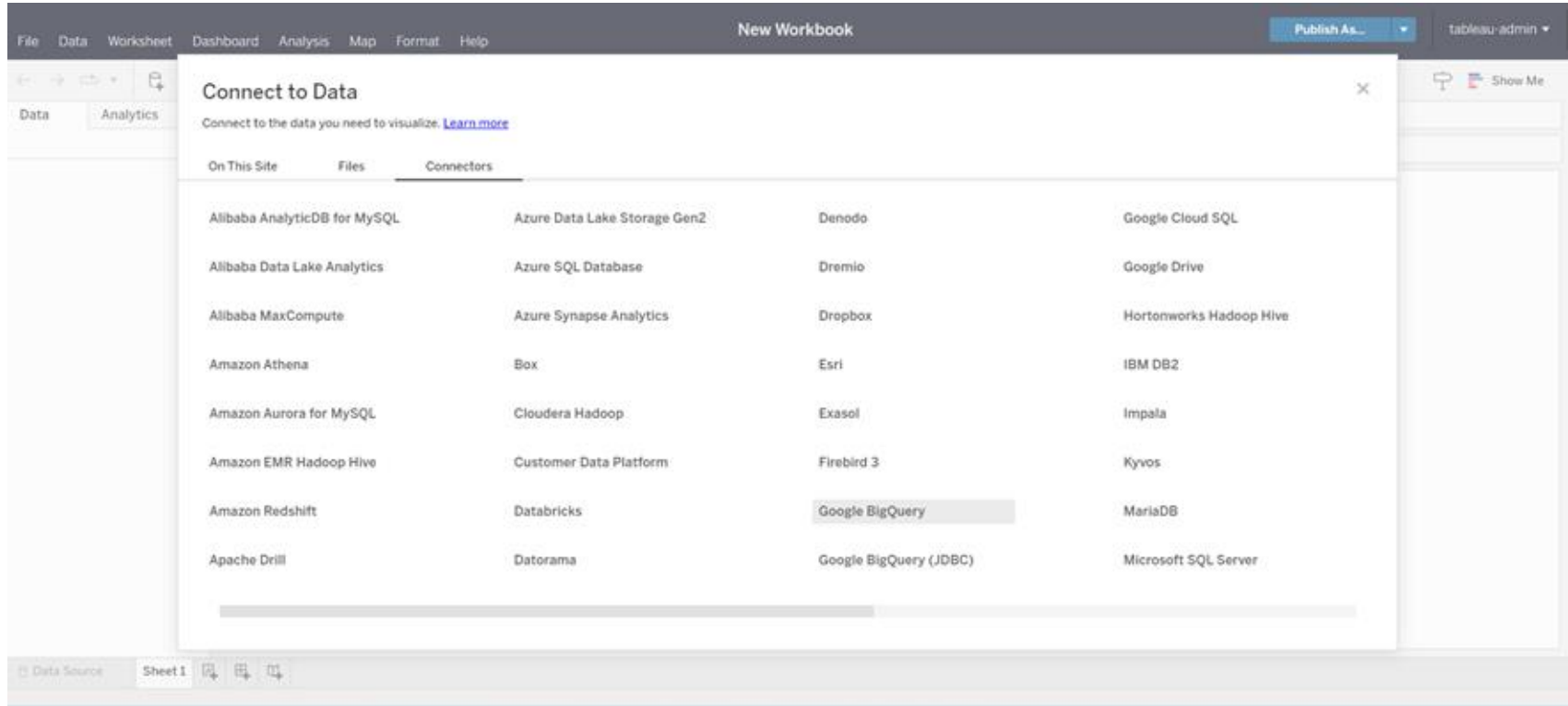
Retrieve the necessary credentials for OAuth connections.

4 . Add OAuth Clients Registry on Tableau Server



Register OAuth clients on Tableau Server for secure data access.

5 . Testing Connection to BigQuery



Verify the connection to BigQuery to ensure successful integration.

6 . Confirm BigQuery Connection

The screenshot displays the Tableau 'New Workbook' interface. On the left sidebar, the 'Connections' section shows 'BigQuery' (Google BigQuery) as the selected connection. Below it, the 'Billing Project' is set to 'My First Project', the 'Project' is 'publicdata', and the 'Dataset' is 'samples'. The 'Table' section lists available tables: 'shakespeare', 'trigrams', 'wikipedia', 'New Custom SQL', 'New Union', 'Use Legacy SQL', and 'New Table Extension'. The main workspace shows a 'wikipedia (samples)' connection with a 'wikipedia' table selected. A message asks 'Need more data? Drag tables here to relate them. [Learn more](#)'. Below this, a table preview is shown with columns: Name, Title, Id, Language, Wp Namespace, Is Redirect, Revision Id, Contributor Ip, and Con. The table contains one row of sample data. At the bottom right, there are buttons for 'Update Now' and 'Update Automatically'.

Connections

- BigQuery (Google BigQuery)

Billing Project: My First Project

Project: publicdata

Dataset: samples

Table

- shakespeare
- trigrams
- wikipedia
- New Custom SQL
- New Union
- Use Legacy SQL
- New Table Extension

wikipedia (samples)

wikipedia

Need more data? Drag tables here to relate them. [Learn more](#)

Name	Title	Id	Language	Wp Namespace	Is Redirect	Revision Id	Contributor Ip	Con
wikipedia	wikipedia	wikipedia	wikipedia	wikipedia	wikipedia	wikipedia	wikipedia	wikipedia

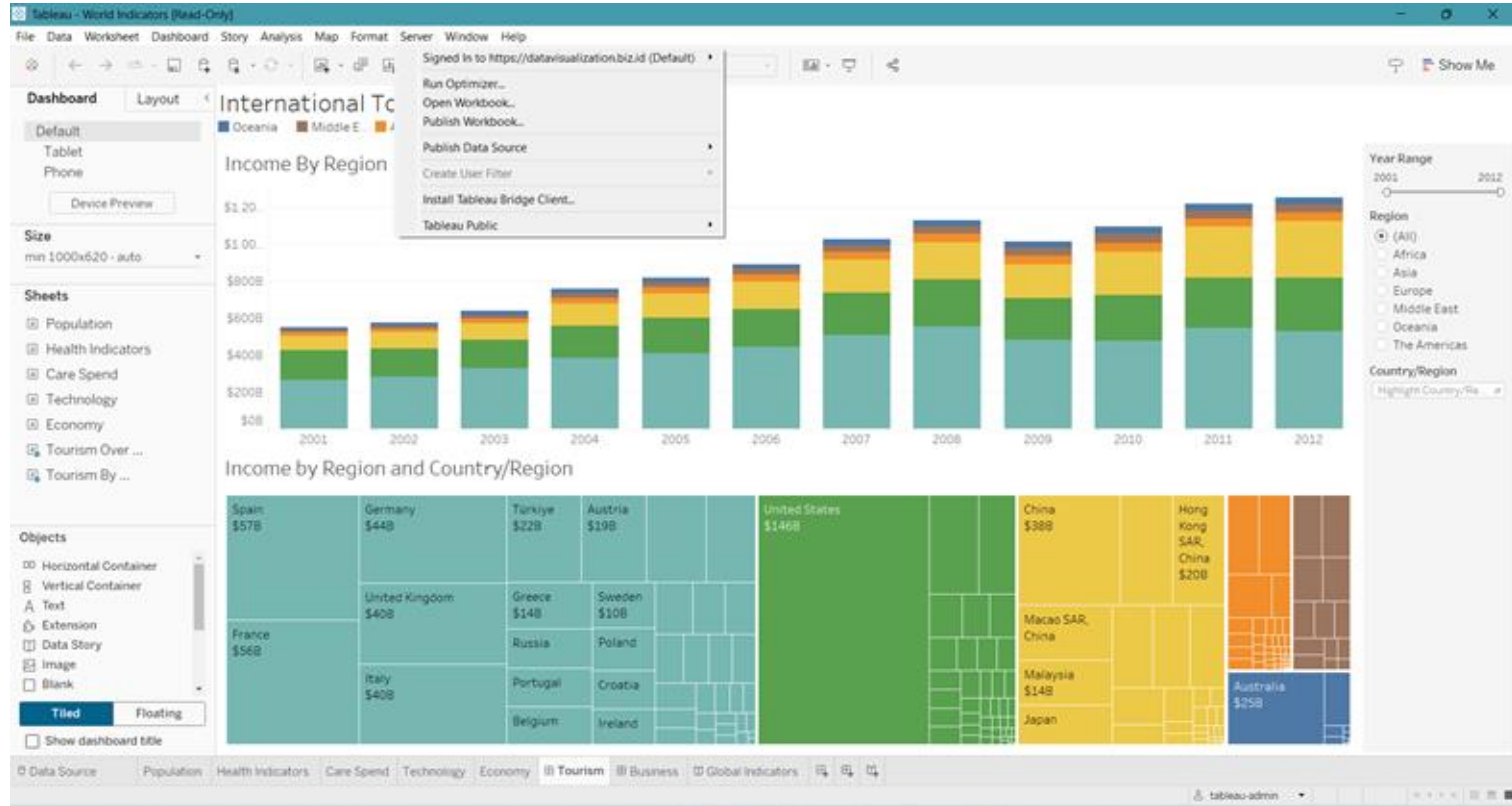
Update Now

Update Automatically

Ensure that the connection to BigQuery is functioning as expected.

Testing Publish Tableau Workbook

1 . Open Tableau Desktop



Launch Tableau Desktop to prepare for publishing.

2 . Publish Dashboard

Tableau - World Indicators [Read-Only]

File Data Worksheet Dashboard Story Analysis Map Format Server Window Help

Dashboard Layout

Default
Tablet
Phone

Device Preview

Size
min 1000x620 - auto

Sheets

- Population
- Health Indicators
- Care Spend
- Technology
- Economy
- Tourism Over ...
- Tourism By ...

Objects

- Horizontal Container
- Vertical Container
- Text
- Extension
- Data Story
- Image
- Blank

Tiled Floating

Show dashboard title

International Tourism

Income By Region

2001 2002 2003

Income by Region and Country/Region

2009 2010 2011 2012

Publish Workbook to Tableau Server

Location: Default

Name: World Indicators - Testing

Description:

Tags: Add

Sheets: All Edit

Permissions: Same as project (Default) Edit

Data Sources: 1 embedded in workbook Edit

More Options:

- Show sheets as tabs
- Show selections
- Include external files

Warning: Server is running an older version than your copy of Tableau Desktop. The workbook will be upgraded on the server and may lose some functionality. All data sources must be embedded in the workbook.

Workbook Optimizer

Year Range: 2001 2012

Region: (All) Africa Asia Europe Middle East Oceania The Americas

Country/Region: Highlight Country/Region

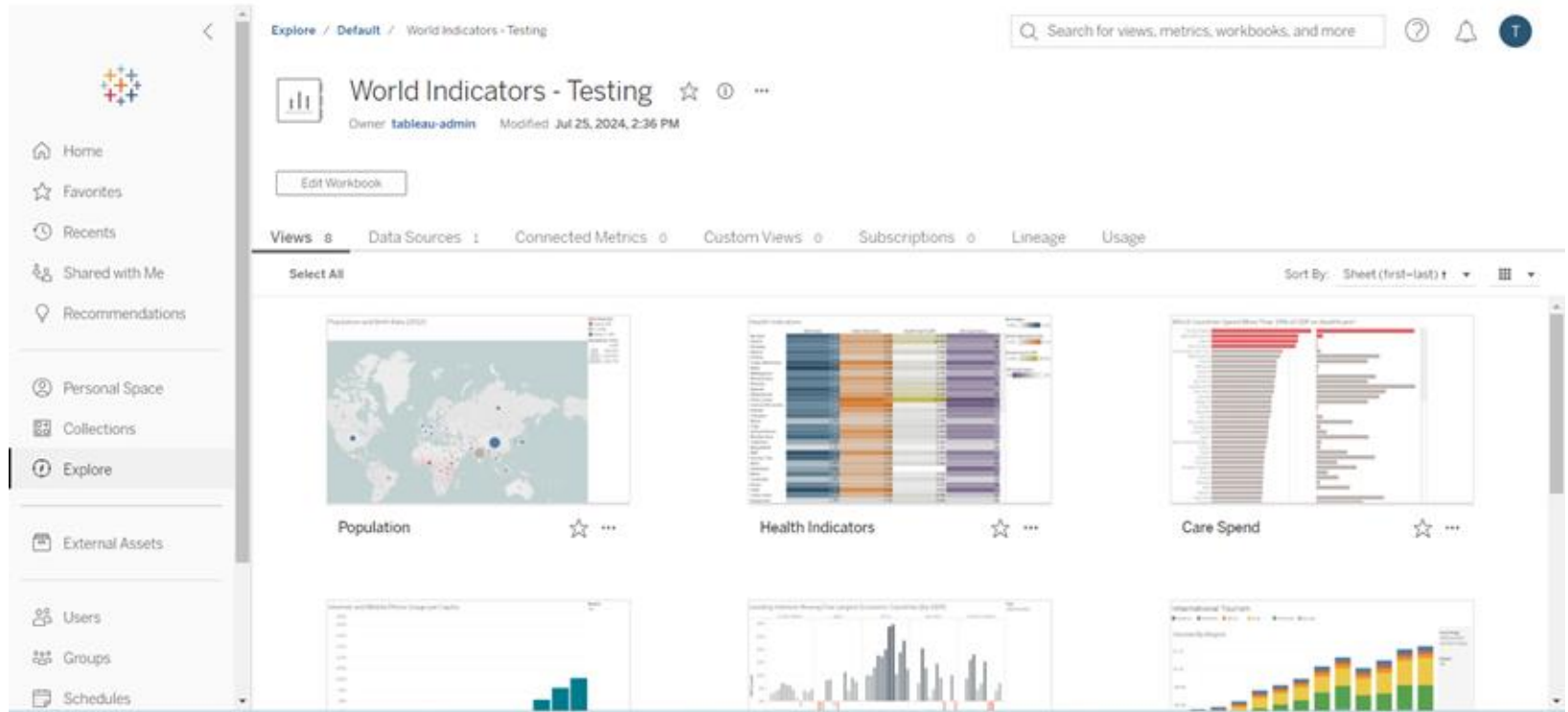
China \$388 Hong Kong SAR, China \$208 Macao SAR, China Malaysia \$148 Japan Australia \$258

Spain \$578 Germany \$448 United Kingdom \$408 France \$568 Italy \$408

Tableau Desktop interface showing the 'Publish Workbook to Tableau Server' dialog box. The dialog box is open, allowing the user to configure the publication settings for the 'World Indicators - Testing' workbook. The background shows a Tableau dashboard with two charts: 'Income By Region' (a stacked bar chart for years 2001-2003) and 'Income by Region and Country/Region' (a treemap for years 2009-2012). The dialog box includes fields for Location, Name, Description, Tags, Sheets, Permissions, and Data Sources. It also has checkboxes for 'Show sheets as tabs', 'Show selections', and 'Include external files'. A 'Warning' message is displayed at the bottom of the dialog, indicating that the server is running an older version of Tableau Desktop and the workbook will be upgraded on the server, potentially losing some functionality. The 'Publish' button is visible at the bottom right of the dialog box.

Execute the process to publish your dashboard from Tableau Desktop to the Tableau Server.

3 . Dashboard Has Been Published



Verify that the dashboard has been successfully published.

References

- <https://help.tableau.com/current/server-linux/en-us/setup.htm>
- https://help.tableau.com/current/offline/en-us/tableau_server_enterprise_deployment_guide.pdf
- <https://www.tableau.com/support/releases/server>
- https://help.tableau.com/current/server/en-us/server_hardware_min.htm
- <https://www.tableau.com/support/drivers>
- https://help.tableau.com/current/server-linux/en-us/startup.htm?_gl=1*1a0hdx7*_ga*MjA4MTc5OTMxNS4xNjk4OTEwNTYz*_ga_8YLN0SNXVS*MTcyMTg3OTkxMy45Ni4xLjE3MjE4ODE1NzguMC4wLjA
- <https://www.tableau.com/learn/whitepapers/google-bigquery-tableau-best-practices#form>
- <https://help.tableau.com/current/server/en-us/ports.htm>
- https://help.tableau.com/current/desktopdeploy/en-us/desktop_deploy_proxy.htm
- <https://www.tableau.com/support/drivers>
- <https://cloud.google.com/dns/docs/set-up-dns-records-domain-name>
- <https://www.youtube.com/watch?v=g1KKI7JVJH8>
- https://help.tableau.com/current/server/en-us/ssl_config.htm
- <https://www.domainesia.com/>

Thank You